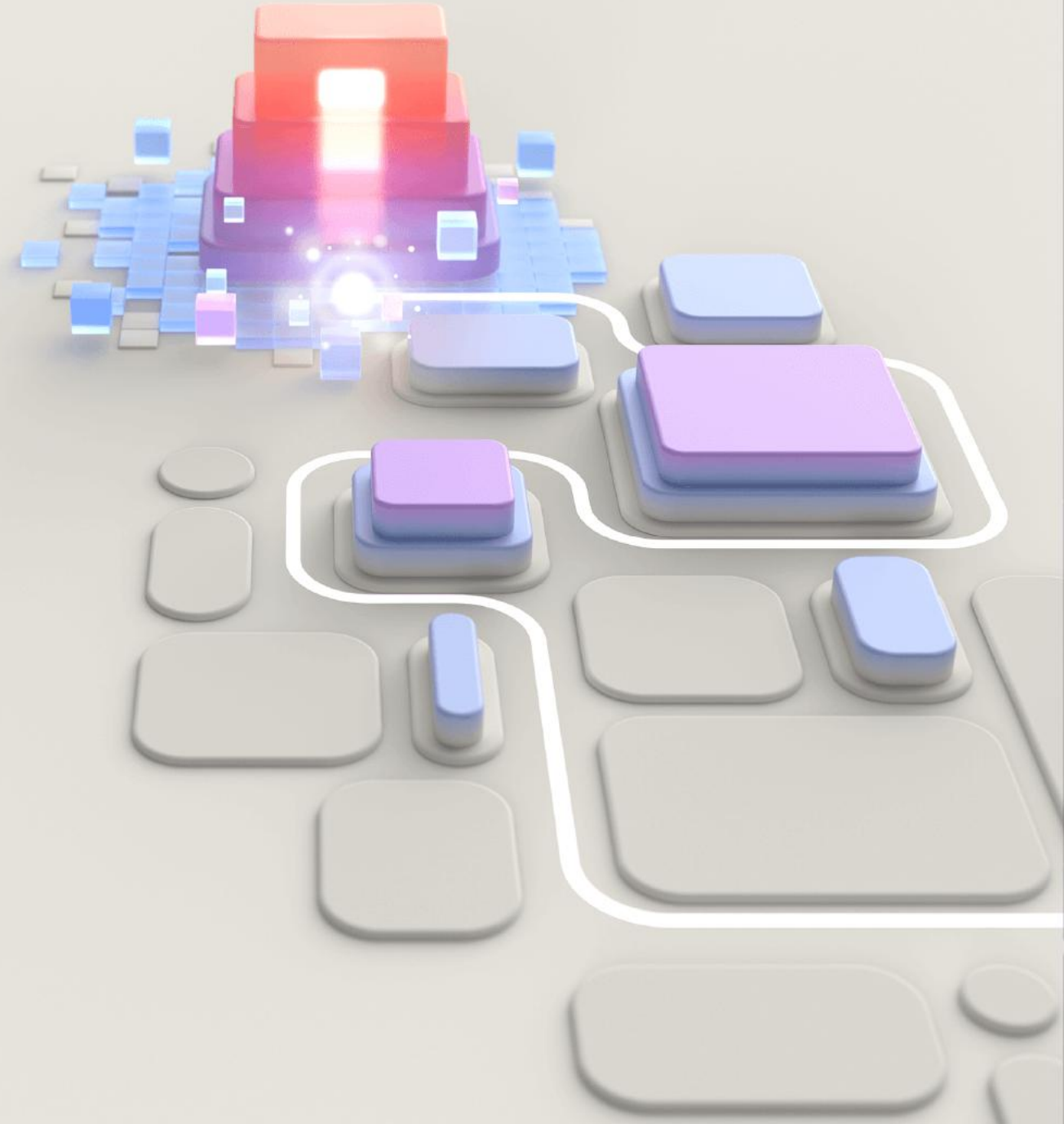




LP 1

Implement an identity management solution

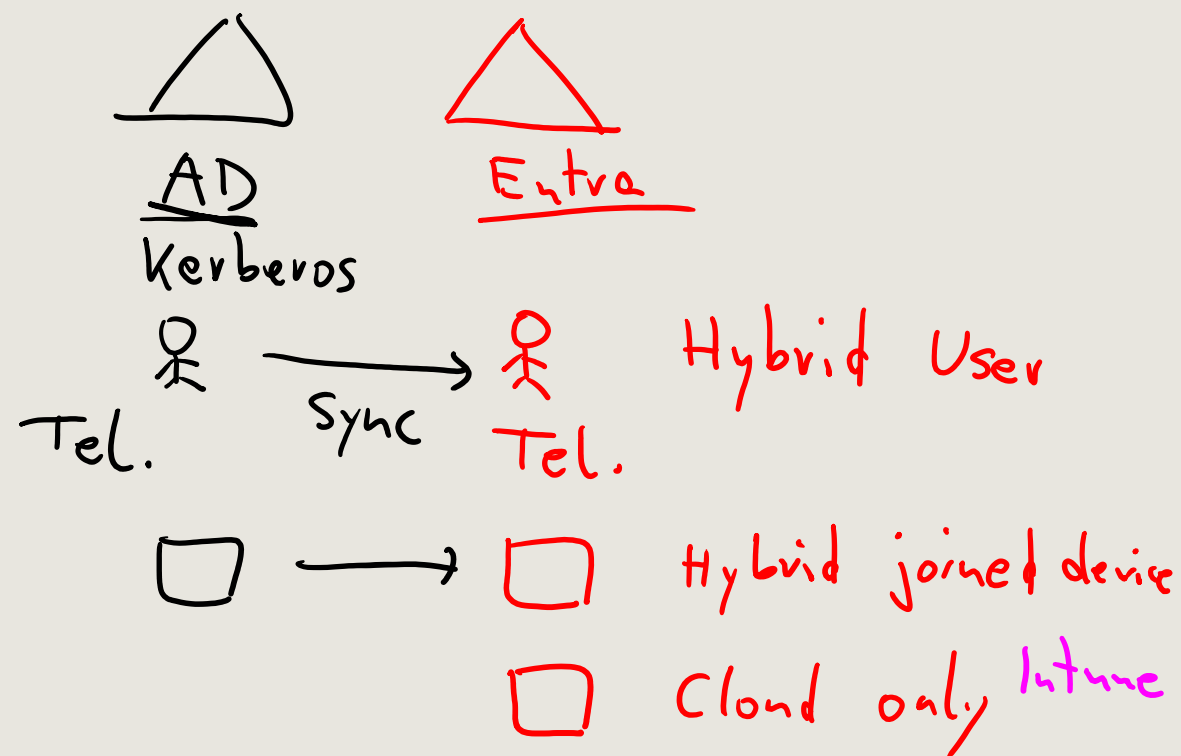
Agenda



- LP 1 Implement an Identity Management Solution
- LP 2 Implement an Authentication and Access Management Solution
- LP 3 Implement Access Management for Apps
- LP 4 Plan and Implement an Identity Governance Strategy

Outline

- Create, configure, and manage identities
- Configure and manage Microsoft Entra ID
- Implement and manage external identities
- Implement and manage hybrid identity



Learning objectives

After completing this module, you will be able to:

- 1** Explain and configure identities for use in your cloud solutions.
- 2** Deploy and manage a Microsoft Entra ID infrastructure.
- 3** Configure and maintain external and hybrid identities.

Create, configure, and manage identities



Create, configure, and manage users

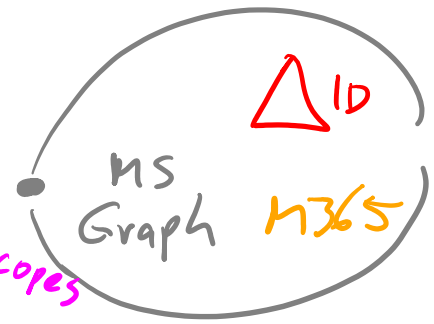


Create, configure, and manage users

- A user account contains all the information needed to authenticate the user during the sign-on process
- You use the **Identity – All users** dashboard in the Microsoft Entra admin center to work with user objects
- Three kinds of users:
 - Cloud identities
 - Directory-synchronized identities
 - External users

PS
Microsoft Graph
Microsoft Entra

Azure Portal
User dev APP
Entra Portal
Admin Center
Powershell Modules



Scopes



The screenshot shows the 'Users' page in the Microsoft Entra Admin Center. The left sidebar contains navigation links: 'All users', 'Audit logs', 'Sign-in logs', 'Diagnose and solve problems', 'Deleted users', 'Password reset', 'User settings', 'Bulk operation results', and 'New support request'. The main content area displays a list of 24 users found. At the top of the main area, there is a search bar, a filter button, and a message stating 'Azure Active Directory is now Microsoft Entra ID.' Below this, a table lists user details.

	Display name ↑	User principal name ↑↓	User type	On-premises sy...
☐	Adele Vance	AdeleV	Member	No
☐	Alex Wilber	AlexW	Member	No
☐	Bhogeswar Kalita	BhogeswarK	Member	No
☐	Diego Siciliani	DiegoS	Member	No
☐	Grady Archie	GradyA	Member	No
☐	Henrietta Mueller	HenriettaM	Member	No
☐	Isaiah Langer	IsaiahL	Member	No
☐	Johanna Lorenz	JohannaL	Member	No

User settings

Some configurable user settings:

- Groups
- Licenses
- Devices
- Location

 **Adele Vance** ...

User

<<

 Overview

 Audit logs

 Sign-in logs

 Diagnose and solve problems

 Custom security attributes

 Assigned roles

 Administrative units

 Groups

 Applications

 Licenses

 Devices

 Azure role assignments

 Authentication methods

 Edit properties

 Delete

 Refresh

 Reset password

 Revoke sessions

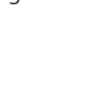
 Manage view

Overview

Monitoring

Properties

Basic info



Adele Vance

AdeleV

Member

User principal name

AdeleV



Object ID

Created date time

Apr 24, 2022, 11:56 AM

User type

Member

Identities

Group memberships

7

Applications

1

Assigned roles

0

Assigned licenses

1

Create, configure, and manage groups



Create, configure, and manage groups

Security groups:

- Most common
- Manage access to shared resources for a group
- Can be nested groups

Microsoft 365 groups:

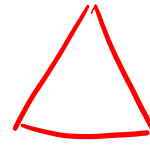
- Access shared mailbox, calendar, SharePoint, and more
- Give access to external people
- Unable to nest within groups

The screenshot displays the 'Groups | All groups' management page. The left sidebar contains navigation options: Overview, All groups (selected), Deleted groups, Diagnose and solve problems, Settings (General, Expiration, Naming policy), Activity (Privileged Identity Management, Access reviews), and a search bar. The main content area shows a list of 24 groups found. The table has columns for Name, Group type, and Membership type. An orange arrow points to the 'Membership type' column, and the word 'Dynamic' is handwritten in orange at the bottom right.

Name	Group type	Membership type
AAD DC Administrators	Security	Assigned
All Company	Microsoft 365	Assigned
Azure ATP	Security	Assigned
Azure ATP	Security	Assigned

Dynamic groups

P1



SKU

- Special type of security group
- Membership dynamically generated via membership rule
 - Property -- example = department, region, and other items

Dynamic membership rules

[Save](#) [Discard](#) | [Got feedback?](#)

[Configure Rules](#) [Validate Rules](#)

You can use the rule builder or rule syntax text box to create or edit a dynamic membership rule. [Learn more](#)

And/Or	Property	Operator	Value
	<Choose a Property>	<Choose an Operator>	Add a value
And	objectId	Not Equals	null
And	Choose a Property	Choose an Operator	Add a value

[+ Add expression](#) [+ Get custom extension properties](#)

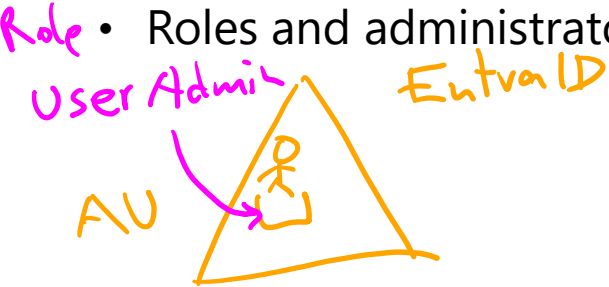
Rule syntax

(user.objectId -ne null)

Group configuration options

Some configurable group settings:

- Properties
- Administrative units
- Group membership
- Roles and administrators



Home > Users > Groups | All groups >

All Company
Group

Overview
Diagnose and solve problems

Manage

- Properties**
- Members
- Owners
- Roles and administrators
- Administrative units**
- Group memberships**
- Applications
- Azure role assignments

Activity

- Privileged Identity Management
- Access reviews

Delete | Got feedback?

Overview

Basic information

All Company

This is the default group for everyone in the network

Membership type	Assigned	Total direct members	1
Source	Cloud	User(s)	1
Type	Microsoft 365	Group(s)	0
Object ID		Device(s)	0
Created on	4/24/2022, 7:51 AM	Other(s)	0
Email	allcompany		

Configure and manage device identities



Microsoft Entra ID registered devices

- Supports “BYOD” (bring your own device)
- Registered devices sign in using a local account
- Also attached to a Microsoft Entra ID account granting access to organizational resources
- Control using Mobile Device Management (MDM) tools like Microsoft Intune

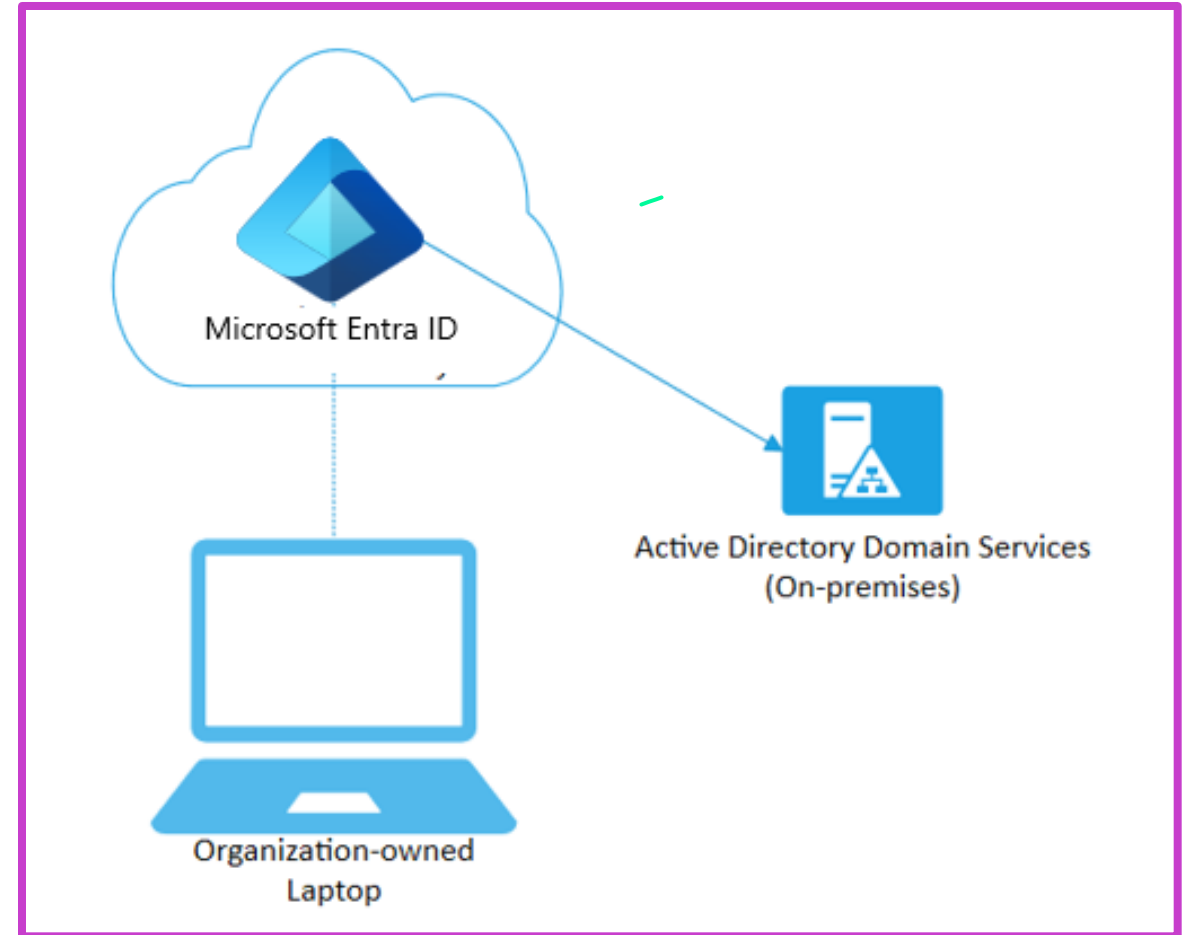
OS: Windows 10/11, iOS, Android, and MacOS



Microsoft Entra ID joined devices ✓

- Intended for cloud-first or cloud-only organizations
- Organization-owned devices
- Joined only to Microsoft Entra ID; organizational account required to sign in
- Easy to sign in with an Entra ID account
- Conditional Access policies can be applied to the device identity

OS: Windows 10/11 devices (not Home)

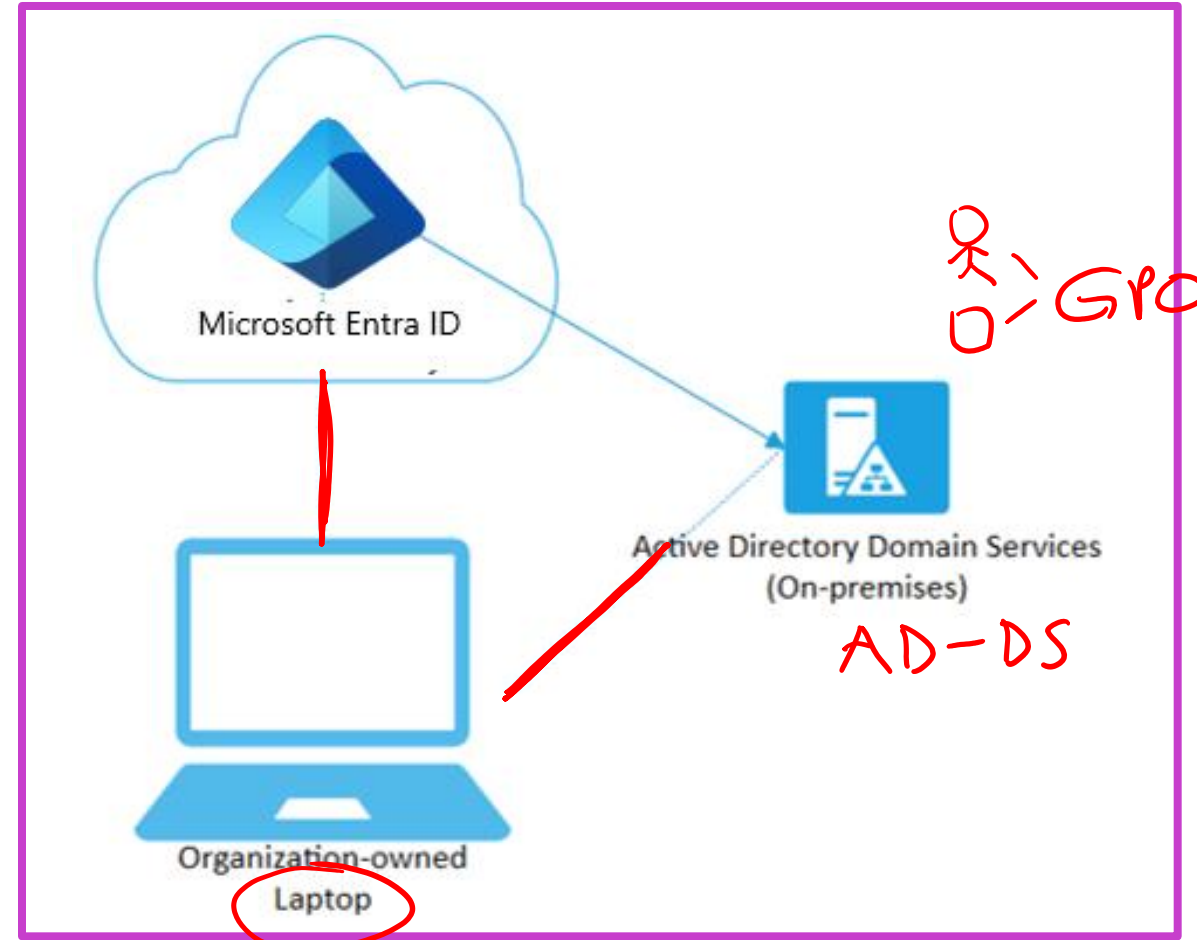


Microsoft Entra hybrid joined devices ~~X~~

Use Microsoft Entra hybrid joined devices if:

- You have Win32 apps deployed to these devices using Microsoft Entra ID machine authentication.
- You want to continue to use group policy to manage the device.
- You want to use existing image solutions to deploy devices.

OS: Windows 8.1 devices in addition to Windows 10/11, plus later Windows Server versions.



Device writeback

- Use Microsoft Entra Connect to copy a cloud registered device to the on-premises AD.
 - Copied into the **Registered Devices** container
- Used to enable device-based conditional access for ADFS-protected devices
- Provides extra security and assurance that access to applications is granted only to trusted devices
- Synchronizes all devices registered in Azure back to the on-premises Active Directory
- Required to Single sign-on to be enabled.

Manage licenses



About licenses

Prof

E3 E5 E7

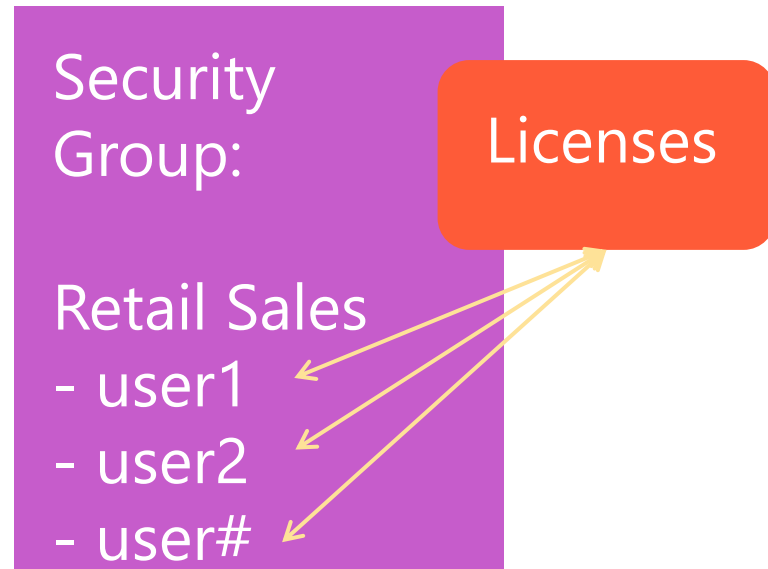
Microsoft Azure is a cloud service that provides many built-in services for free.

- Microsoft Entra ID comes as a free service
- Gain additional Microsoft Entra ID functionality with a premium P1 or P2 license
- Microsoft Entra ID Governance license
 - A targeted governance license focused on entitlement related features, lifecycle workflows, access reviews, and PIM.

Additional services (like O365 are paid cloud services)

- Microsoft paid cloud services require licenses
- Licenses are assigned to each user who needs access to the services
- Each user requires a separate paid license
- Administrators use management portals and PowerShell cmdlets to manage licenses

Group-based licensing



- You can assign one or more product licenses to a group
- Microsoft Entra ID assigns to all members of the group
- New group members are automatically assigned the appropriate licenses
- Licenses are removed from users when they leave the group
- Licenses can be assigned to any security group

Custom security attributes



Manage custom security attributes

Business-specific attributes (key-value pairs) that you can define and assign to Microsoft Entra ID objects.

Example: Add app1 storage clearance attribute to users who need access; then set a conditional access policy.

Microsoft Entra ID P1/P2

Custom attributes

+ Add attribute set

Refresh

Got feedback?

Search attribute set name

Attribute set name	Description	Maximum number of attributes
Banker	This person can write checks against the bank	25
Clerk	Bank clerk	25

New attribute set

Add an attribute set to group and manage related custom security attributes. All custom security attributes must be a part of an attribute set.

Attribute set name

Description

Maximum number of attributes

25

New attribute

Add a custom security attribute (key-value pair) to your directory that you can later assign to Microsoft Entra objects, such as users or applications.

Attribute name

Description

Data type

String

Allow multiple values to be assigned

Only allow predefined values to be assigned

Predefined values

No

No

+ Add value

Value	Is active?
No results	

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Microsoft Entra Agent ID and Entra Embedded Agents

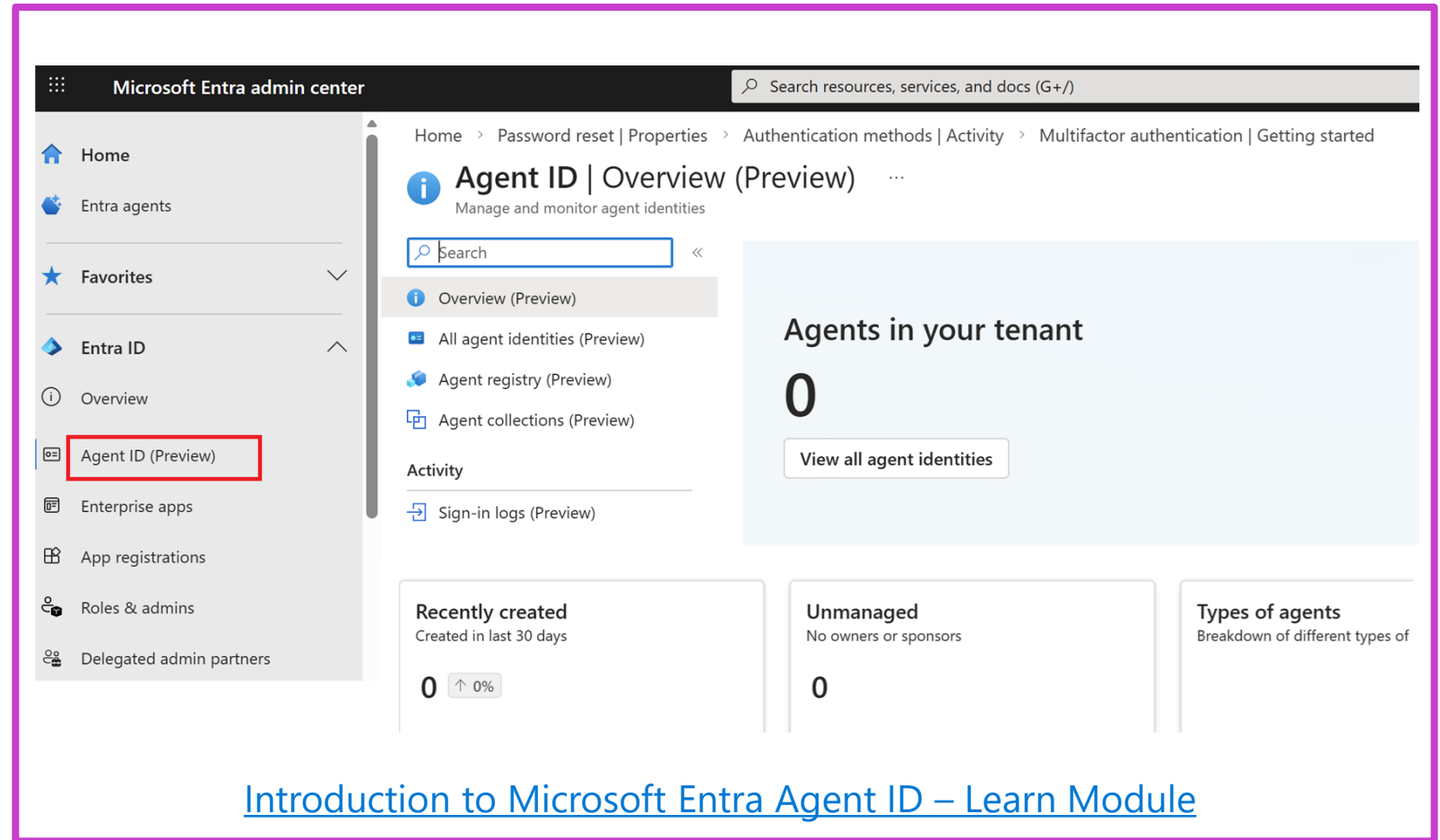


Microsoft Entra Agent ID (preview)

New type of Identity

Agents need identities

All the Entra tools from
Conditional Access to
Entitlement work with
Agent IDs.



Microsoft Entra Embedded Agents

**Conditional Access
Optimization Agent**

Access Review Agent

**Identity Risk Management
Agent**

App Registration Agent

The agents are in preview and are not guaranteed to move to GA. Each is exploring a possible use of AI and Security Copilot to help you secure your environment. Please feel free to experiment and provide feedback on all of them.

- Security Copilot with SCU's required to use these agents.

Configure and manage a Microsoft Entra tenant



Objectives

- 1 Configure company branding
- 2 Configure and manage Microsoft Entra roles
- 3 Configure delegation by using administrative units
- 4 Evaluate effective permissions
- 5 Configure and manage domains in Microsoft Entra and Microsoft 365
- 6 Configure tenant-wide settings

Company branding



Company branding

Customize the Microsoft Entra ID sign-in pages with company logos and other elements

- Requires Microsoft Entra ID P1/P2, or an M365 license
- Customize by language

Customize default sign-in experience ...

[Basics](#) [Layout](#) [Header](#) [Footer](#) [Sign-in form](#) [Review](#)

Customize the default sign-in experience to personalize the sign-in page for anyone signing-in to your tenant.

Background

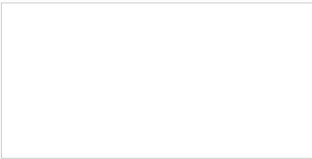
Upload background image, favicon or choose background color.

Favicon ⓘ

 Image size: 32x32px (resizable)
Max file size: 5KB
File Type: PNG (preferred), JPG, or JPEG

Background image ⓘ

ⓘ Image will appear darkened to improve contrast and legibility.

 Image size: 1920x1080px
Max file size: 300KB
File Type: PNG, JPG, or JPEG

Page background color ⓘ Not provided



Configure and manage Microsoft Entra roles



Admin access to Azure and Microsoft Entra ID

Entra Admin Center—<https://entra.microsoft.com>

Azure portal—<https://portal.azure.com>

M365 Admin Center—<https://admin.microsoft.com>

Microsoft Defender for Cloud Apps Portal—<https://security.microsoft.com/>

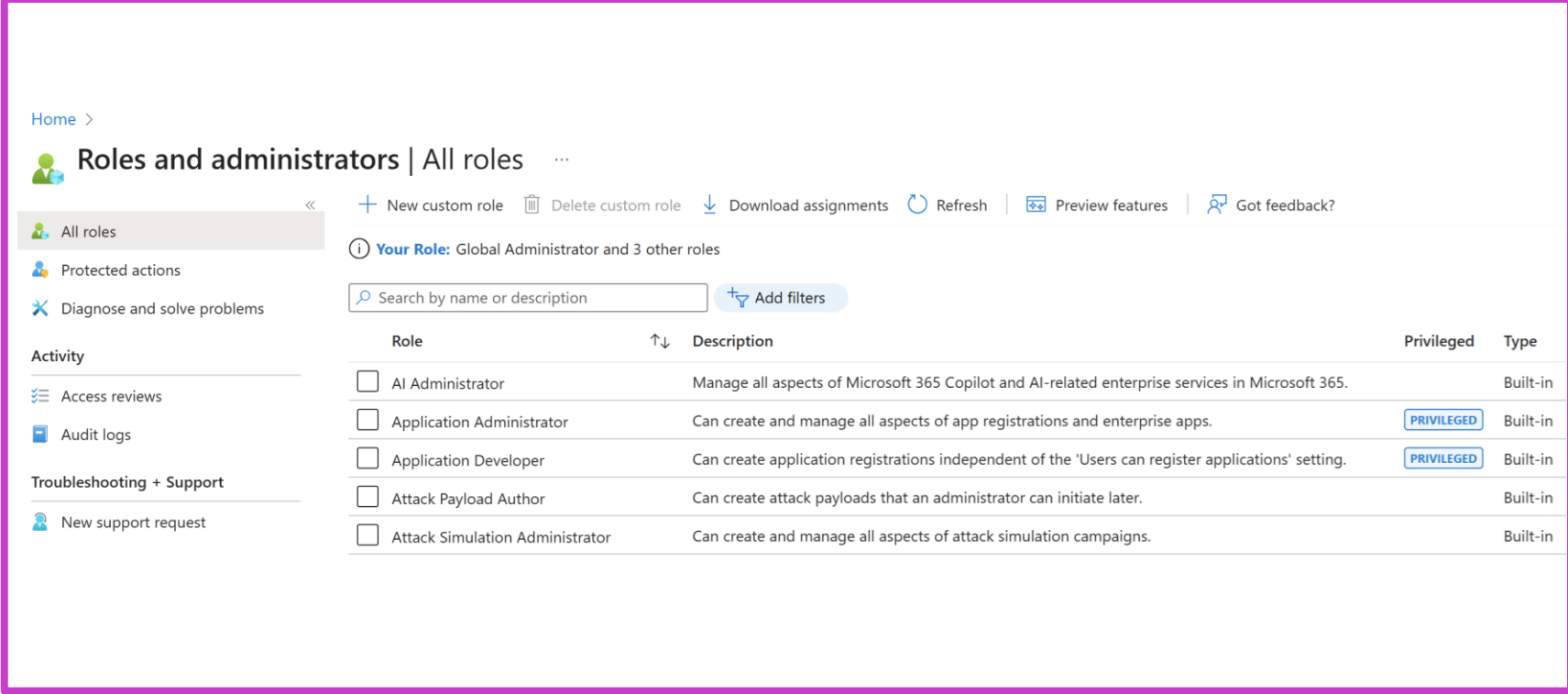
Microsoft Entra roles

Intended for:

- IT admins
- App developers
- Microsoft 365, Office 365, Azure, or Dynamics CRM Online subscribers

Define access to Entra ID resources

Do not overlap with Azure roles by default



The screenshot displays the 'Roles and administrators' page in the Microsoft Entra admin center. The left sidebar contains navigation links: 'All roles' (selected), 'Protected actions', 'Diagnose and solve problems', 'Activity', 'Access reviews', 'Audit logs', 'Troubleshooting + Support', and 'New support request'. The main content area shows a list of roles with columns for 'Role', 'Description', 'Privileged', and 'Type'. The roles listed are AI Administrator, Application Administrator, Application Developer, Attack Payload Author, and Attack Simulation Administrator. The first three roles are marked as 'Privileged' and 'Built-in'. The interface also includes a search bar, a filter button, and a 'Your Role' indicator showing 'Global Administrator and 3 other roles'.

Role	Description	Privileged	Type
☐ AI Administrator	Manage all aspects of Microsoft 365 Copilot and AI-related enterprise services in Microsoft 365.		Built-in
☐ Application Administrator	Can create and manage all aspects of app registrations and enterprise apps.	PRIVILEGED	Built-in
☐ Application Developer	Can create application registrations independent of the 'Users can register applications' setting.	PRIVILEGED	Built-in
☐ Attack Payload Author	Can create attack payloads that an administrator can initiate later.		Built-in
☐ Attack Simulation Administrator	Can create and manage all aspects of attack simulation campaigns.		Built-in

Roles and administrators

All roles

Protected actions

Diagnose and solve problems

Activity

Access reviews

Audit logs

Troubleshooting + Support

New support request

+ New custom role

Delete custom role

Download assignments

Refresh

Preview features

Got feedback?

Get just-in-time access to a role when you need it using PIM. Learn more about PIM

BasicsPermissionsReview + create

Learn more

Roles created here will be available for assignment on other resources as well. Learn more

Search

Name *

Description

Baseline permissions

Start from scratch

Clone from a custom role

Attack Simulation Administrator

Can create and manage all aspects of attack

Attribute Assignment Administrator

Assign custom security attribute keys and va

Attribute Assignment Reader

Read custom security attribute keys and valu

Attribute Definition Administrator

Define and manage the definition of custom

BasicsPermissionsReview + create

Add permissions for this custom role. Currently, permissions for Application registrations and Enterprise applications are supported in custom roles. Learn more

Search by permission name or description

Permission	Description
microsoft.directory/applicationPolicies/allProperties/read	Read all properties (including privileged properties) on application policies
microsoft.directory/applicationPolicies/allProperties/update	Update all properties (including privileged properties) on application policies
microsoft.directory/applicationPolicies/basic/update	Update standard properties of application policies
microsoft.directory/applicationPolicies/create	Create application policies
microsoft.directory/applicationPolicies/createAsOwner	Create application policies, and creator is added as the first owner

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Requires a Microsoft Entra ID premium license

Assigning roles—to a user or group

Home > Users > Adele Vance

 Adele Vance | Assigned roles

...

User

Search

<<

+ Add assignments

Refresh

Got feedback?

Overview

Audit logs

Sign-in logs

Diagnose and solve problems

Manage

Custom security attributes

Assigned roles

Administrative units

Eligible assignments

Active assignments

Expired assignments

Search by role

Role	↑↓	Principal name	Scope	↑↓	Membership
Application Administrator		AdeleV	Directory		Direct

Assign built-in or custom roles

Assign to a user, group, service principal or managed identity

Use principle of least privilege when assigning

Assign users or groups to a role

The screenshot shows the Azure portal interface for assigning users or groups to a role. The main page is titled 'Billing administrator | Assignments' and includes a sidebar with navigation options like 'Diagnose and solve problems', 'Manage', 'Activity', and 'Troubleshooting + Support'. A green box highlights the '+ Add assignments' button. To the right, a modal window titled 'Add assignments' is open, displaying a search bar and a list of eligible users. The list includes Adele Vance, Alex Wilber, and Allan Deyoung, each with their profile picture, email address, and organization name. Below the list, a 'Selected items' section shows 'No items selected'.

Home > Contoso > Billing administrator

Billing administrator | Assignments
All roles

<< **+ Add assignments** >>

✖ Diagnose and solve problems

Manage

- Assignments
- Description

Activity

- Bulk operation results

Troubleshooting + Support

- New support request

Add assignments ✕

Only groups eligible for role assignment are displayed. [Learn more](#)

Search ⓘ

Search

	Adele Vance	AdeleV@	Contoso	.OnMicrosoft.com
	Alex Wilber	AlexW@	Contoso	OnMicrosoft.com
	Allan Deyoung			

Selected items

No items selected

Identities can be added
to a role

Alternative method to
assign roles

Use principle of least
privilege when
assigning

Exercise—Explore dynamic groups



This exercise teaches students how to create a dynamic group with all users as members.

[Launch this Exercise in GitHub](#)

Dynamic membership rules

 Save  Discard |  Got feedback?

Configure Rules

Validate Rules (Preview)

You can use the rule builder or rule syntax text box to create or edit a dynamic mem

And/Or	Property
And	<Choose a Property>

 Add expression

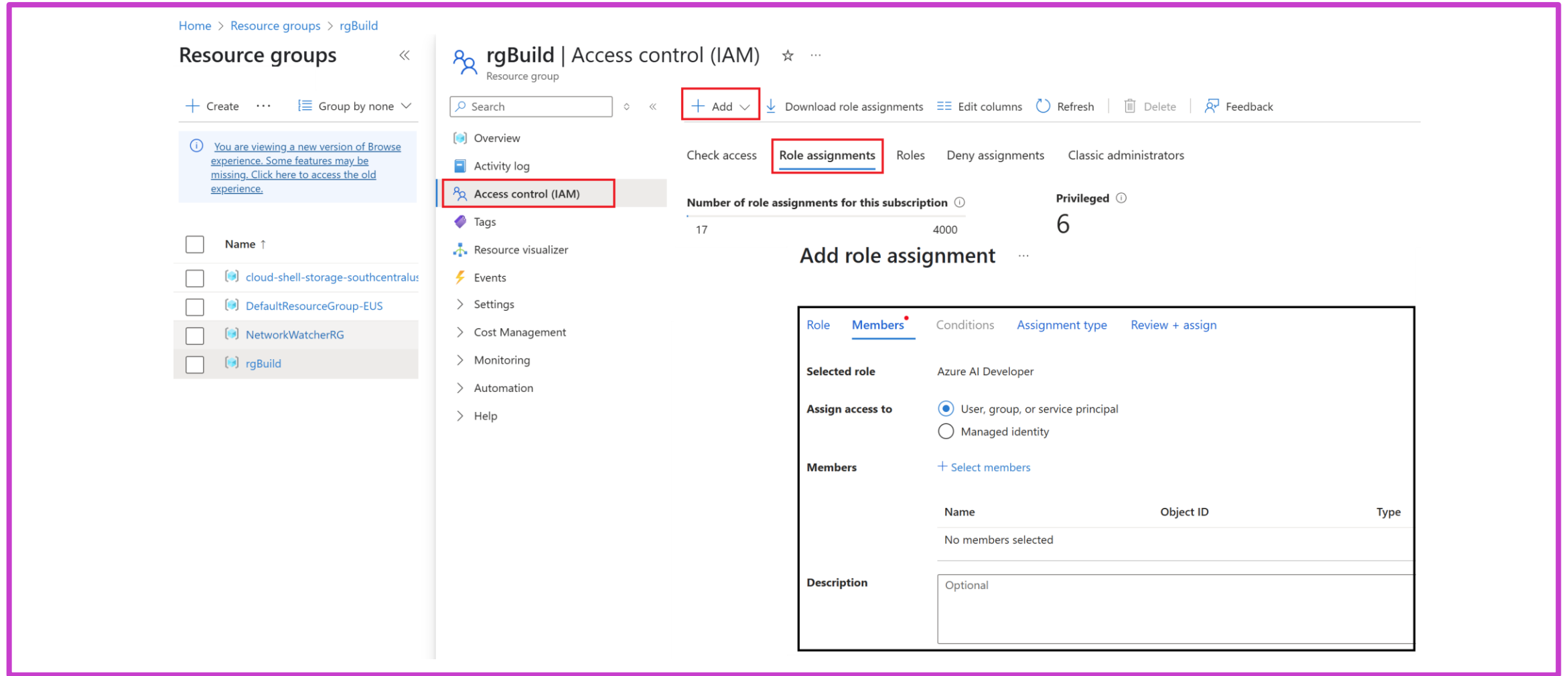
 Get custom extension properties ⓘ

 Some items could not be displayed in the rule builder. [Learn more](#)

Rule syntax

user.objectId -ne null

Assigning Azure roles to a subscription or broader scope



The screenshot displays the Azure portal interface for managing access control (IAM) for the resource group 'rgBuild'. The left sidebar shows the 'Resource groups' list with 'rgBuild' selected. The main pane shows the 'Access control (IAM)' page, where the 'Add' button is highlighted in red. The 'Role assignments' tab is also highlighted in red. Below the tabs, the 'Add role assignment' dialog is open, showing the 'Members' tab. The dialog includes a table for selecting members, with columns for Name, Object ID, and Type. The table currently shows 'No members selected'.

Home > Resource groups > rgBuild

Resource groups <<

+ Create ... Group by none ▾

rgBuild | Access control (IAM) ☆ ...

Resource group

Search ▾ << + Add ▾ Download role assignments Edit columns Refresh Delete Feedback

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Check access **Role assignments** Roles Deny assignments Classic administrators

Number of role assignments for this subscription ⓘ 17 4000

Privileged ⓘ 6

Add role assignment ...

Role **Members** Conditions Assignment type Review + assign

Selected role Azure AI Developer

Assign access to ☒ User, group, or service principal ☐ Managed identity

Members + Select members

Name	Object ID	Type
No members selected		

Description Optional

Assign roles to manage subscriptions, management groups, and resource groups

Using Microsoft Entra roles for Common delegation scenarios

Delegation scenarios:

- Restrict and manage application creation
- Assigning owners to an application
- Assigning a built-in administrative role that grants access to manage configuration in Microsoft Entra ID for all applications
- Creating a custom role defining very specific permissions and assigning it to someone

**When you delegate administrative tasks, you want to keep a few terms in mind:
Least Privilege – Just in Time – Just long enough**

Delegating administrative tasks with built-in roles

Delegating app administration

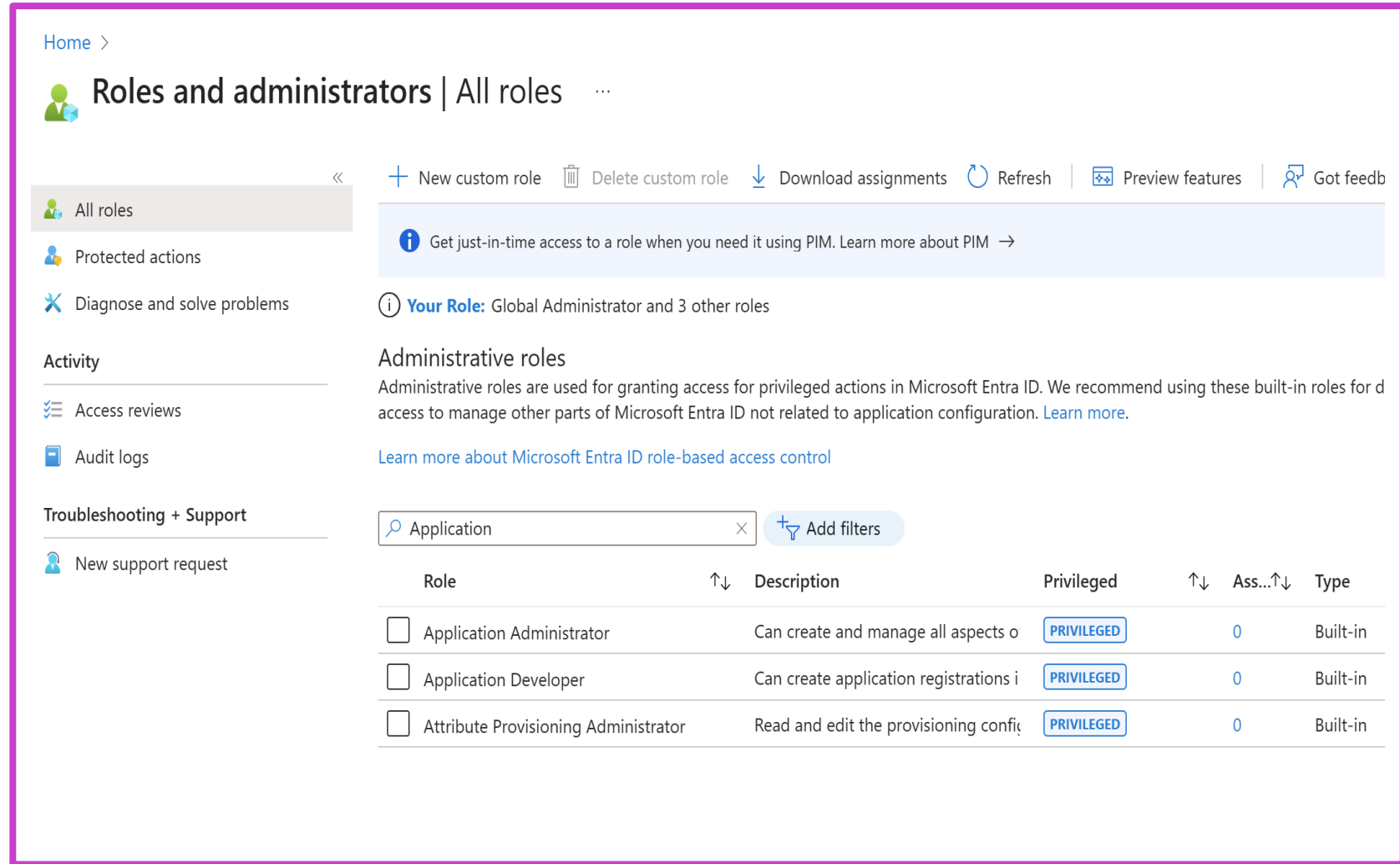
- Password administrator role
- Group administrator role

Delegating app registration

- Application developer role

Delegating app ownership

- Enterprise application owner role
- Application registration owner role



The screenshot displays the 'Roles and administrators' page in the Microsoft Entra ID portal. The left sidebar contains navigation links: 'All roles' (selected), 'Protected actions', 'Diagnose and solve problems', 'Activity' (with sub-links for 'Access reviews' and 'Audit logs'), and 'Troubleshooting + Support' (with a link for 'New support request'). The main content area shows the 'All roles' view with a search bar containing 'Application' and an 'Add filters' button. Below the search bar is a table of administrative roles. The table has columns for 'Role', 'Description', 'Privileged' status, 'Ass...' (assignments), and 'Type'. Three roles are listed: 'Application Administrator', 'Application Developer', and 'Attribute Provisioning Administrator'. All three roles are marked as 'PRIVILEGED' and have 0 assignments. A banner at the top of the main content area provides information about just-in-time access using PIM.

Home > Roles and administrators | All roles ...

+ New custom role | Delete custom role | Download assignments | Refresh | Preview features | Got feedback

Get just-in-time access to a role when you need it using PIM. Learn more about PIM →

Your Role: Global Administrator and 3 other roles

Administrative roles

Administrative roles are used for granting access for privileged actions in Microsoft Entra ID. We recommend using these built-in roles for d access to manage other parts of Microsoft Entra ID not related to application configuration. [Learn more.](#)

[Learn more about Microsoft Entra ID role-based access control](#)

Application

Role	↑↓	Description	Privileged	↑↓	Ass...↑↓	Type
☐ Application Administrator		Can create and manage all aspects o	PRIVILEGED		0	Built-in
☐ Application Developer		Can create application registrations i	PRIVILEGED		0	Built-in
☐ Attribute Provisioning Administrator		Read and edit the provisioning confi	PRIVILEGED		0	Built-in

Demo—assigning roles

Resource type

Directory

Select role ⓘ

Application Administrator

Scope type ⓘ

Directory

Select member(s) * ⓘ

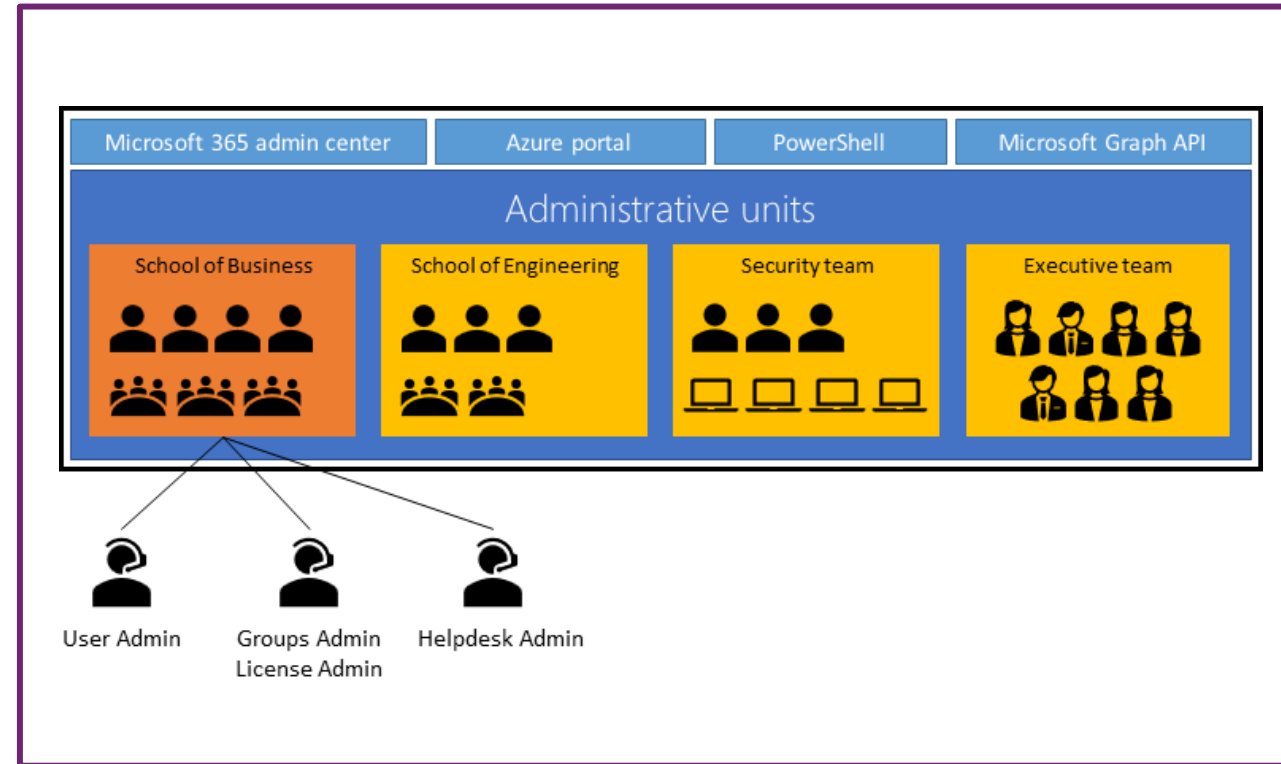
No member selected

When to use Administrative units



About administrative units

- Administrative units are Microsoft Entra ID resources that can be containers for other Microsoft Entra resources.
- An administrative unit can contain only users, groups, and devices.
- Administrative units restrict permissions in a role to a specific portion of your organization that you define.



Deployment scenario

Large university composed of many autonomous schools (School of Business, School of Engineering, and so on)

Each school has a team of IT admins who control access, manage users, and set policies for their school

Administrative tasks could include:

- Creating a role with administrative permissions over only Microsoft Entra users in the business school administrative unit
- Creating an administrative unit for the School of Business
- Populating the administrative unit with only the business school students and staff
- Adding the business school IT team to the role, along with its scope

Common delegation scenarios for Admin Units

Delegation scenarios:

- Assigning types of authentication methods
- Creating a dedicated Helpdesk admin
- Admin to control users and groups within the admin unit
- Set up a person who manages the licenses to tools and SaaS apps

When you delegate administrative tasks, you want to keep a few terms in mind:

Least Privilege – Just in Time – Just long enough

Administrative Units

Delegating authentication administration

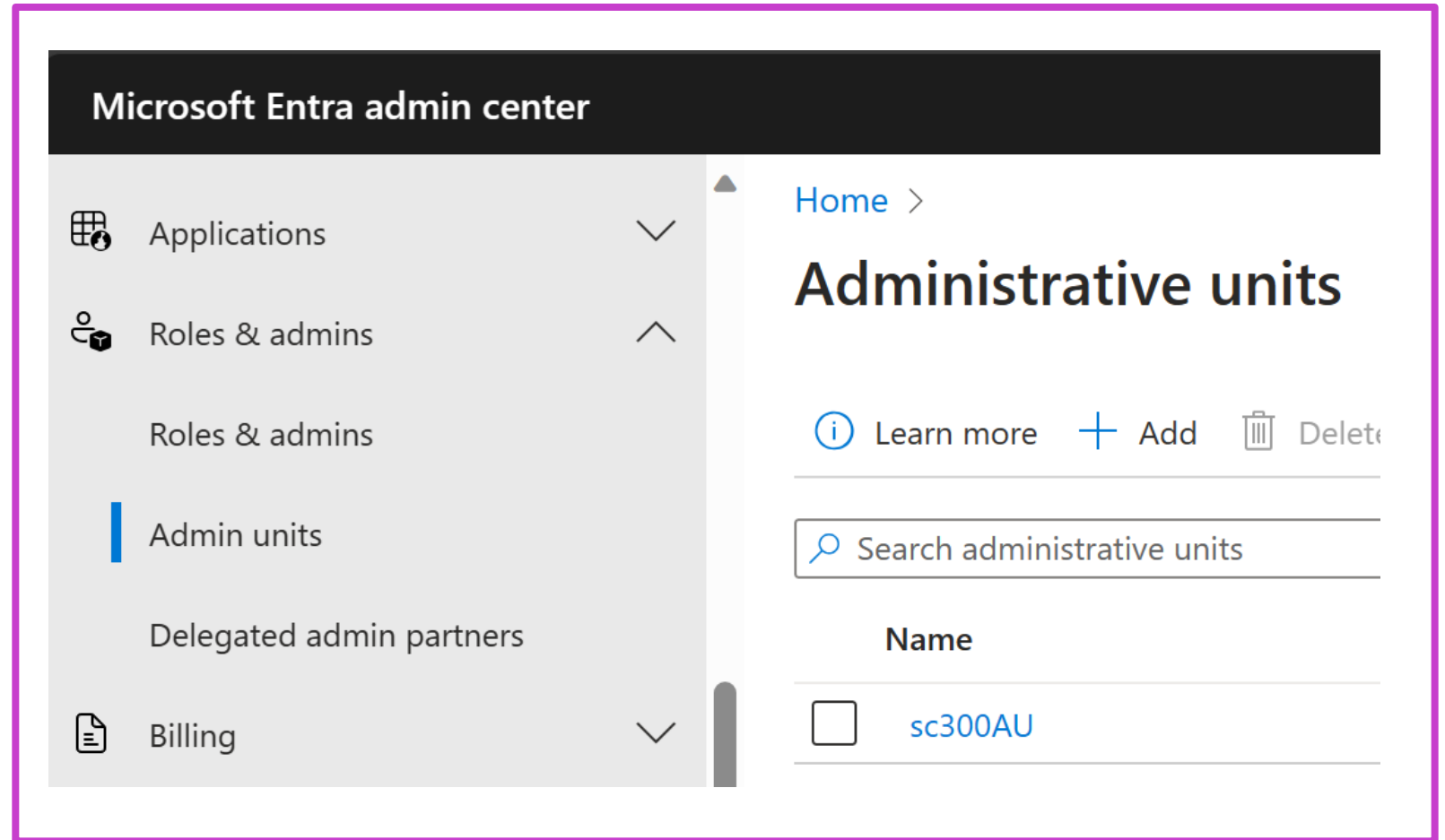
- Authentication administrator role

Delegating licensing

- License administrator

Delegating user, group, and password management

- User administrator



Evaluate effective permissions for Microsoft Entra role



Permissions in Microsoft Entra ID

Permission—consent or authorization to perform a specific action.

Who gets permissions:

Member Users	External Users	Applications	Devices
--------------	----------------	--------------	---------

Example default permissions for each:

• Enumerate list of users • Invite external users • Change their password • Manage Photo • Create groups • And so on	• Read own properties • Change their password • Search for groups • Cannot invite groups • And so on	• App must be registered • API and permissions assigned in registration • Examples – read, write, verify, and others	• Device must be registered or joined to the Microsoft Entra tenant
--	--	--	---

* Augment or restrict permissions with settings and role assignments

Augment or restrict permissions

 **Users | User settings** ×

Default user role permissions

[Learn more](#)

Users can register applications ⓘ

☒ Yes

Restrict non-admin users from creating tenants ⓘ

☐ No

Users can create security groups ⓘ

☒ Yes

Guest user access

[Learn more](#)

Guest user access restrictions ⓘ

☐ Guest users have the same access as members (most inclusive)

☒ Guest users have limited access to properties and memberships of directory objects

☐ Guest user access is restricted to properties and memberships of their own directory objects (most restrictive)

Administration portal

[Learn more](#)

Restrict access to Microsoft Entra ID administration portal ⓘ

☐ No

LinkedIn account connections

[Learn more](#)

Allow users to connect their work or school account with LinkedIn ⓘ *

☒ Yes

☐ Selected group

 **Roles and administrators | All roles** ...

[+ New custom role](#) [Delete custom role](#) [Download assignments](#) [Refresh](#)

 Get just-in-time access to a role when you need it using PIM. [Learn more about PIM](#) →

 **Your Role:** Global Administrator and 3 other roles

Administrative roles

Administrative roles are used for granting access for privileged actions in Microsoft Entra ID. We access to manage other parts of Microsoft Entra ID not related to application configuration. [Learn more about Microsoft Entra ID role-based access control](#)

[Add filters](#)

Role	↑↓	Description
☐ AI Administrator		Manage all aspects of Microsoft 365 Copil
☐ Application Administrator		Can create and manage all aspects of app
☐ Application Developer		Can create application registrations indepe
☐ Attack Payload Author		Can create attack payloads that an admini
☐ Attack Simulation Administrator		Can create and manage all aspects of attac
☐ Attribute Assignment Administrator		Assign custom security attribute keys and
☐ Attribute Assignment Reader		Read custom security attribute keys and v
☐ Attribute Definition Administrator		Define and manage the definition of custo
☐ Attribute Definition Reader		Read the definition of custom security attr

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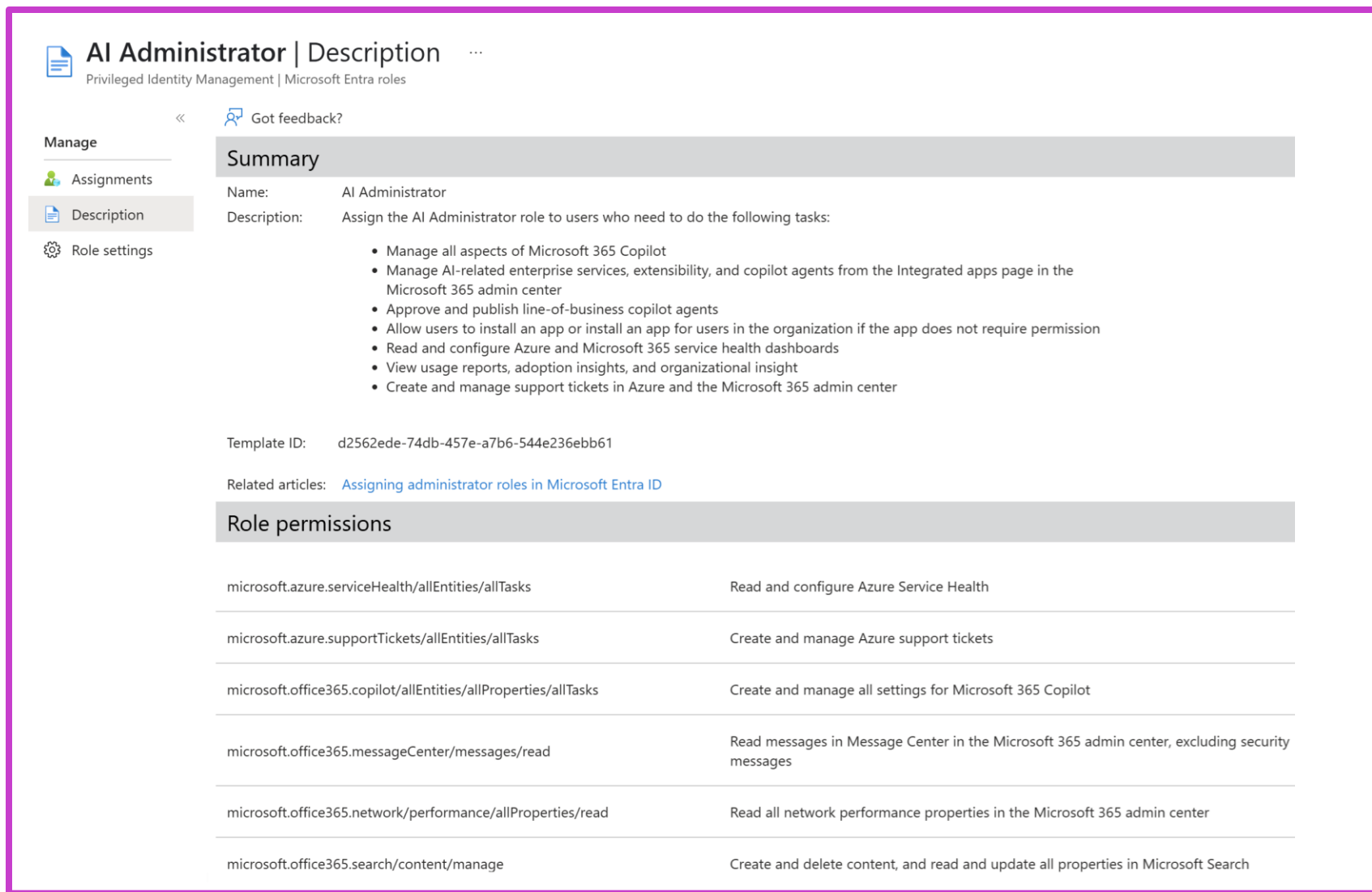
Analyze permission

Role permission

Specific permissions granted when role assigned to a user or group.

Guest/service principal

Additional permission given to ensure a basic level of functionality when assigned to guest users or service principals.



AI Administrator | Description ...
Privileged Identity Management | Microsoft Entra roles

Manage
Assignments
Description
Role settings

Summary

Name: AI Administrator

Description: Assign the AI Administrator role to users who need to do the following tasks:

- Manage all aspects of Microsoft 365 Copilot
- Manage AI-related enterprise services, extensibility, and copilot agents from the Integrated apps page in the Microsoft 365 admin center
- Approve and publish line-of-business copilot agents
- Allow users to install an app or install an app for users in the organization if the app does not require permission
- Read and configure Azure and Microsoft 365 service health dashboards
- View usage reports, adoption insights, and organizational insight
- Create and manage support tickets in Azure and the Microsoft 365 admin center

Template ID: d2562ede-74db-457e-a7b6-544e236ebb61

Related articles: [Assigning administrator roles in Microsoft Entra ID](#)

Role permissions

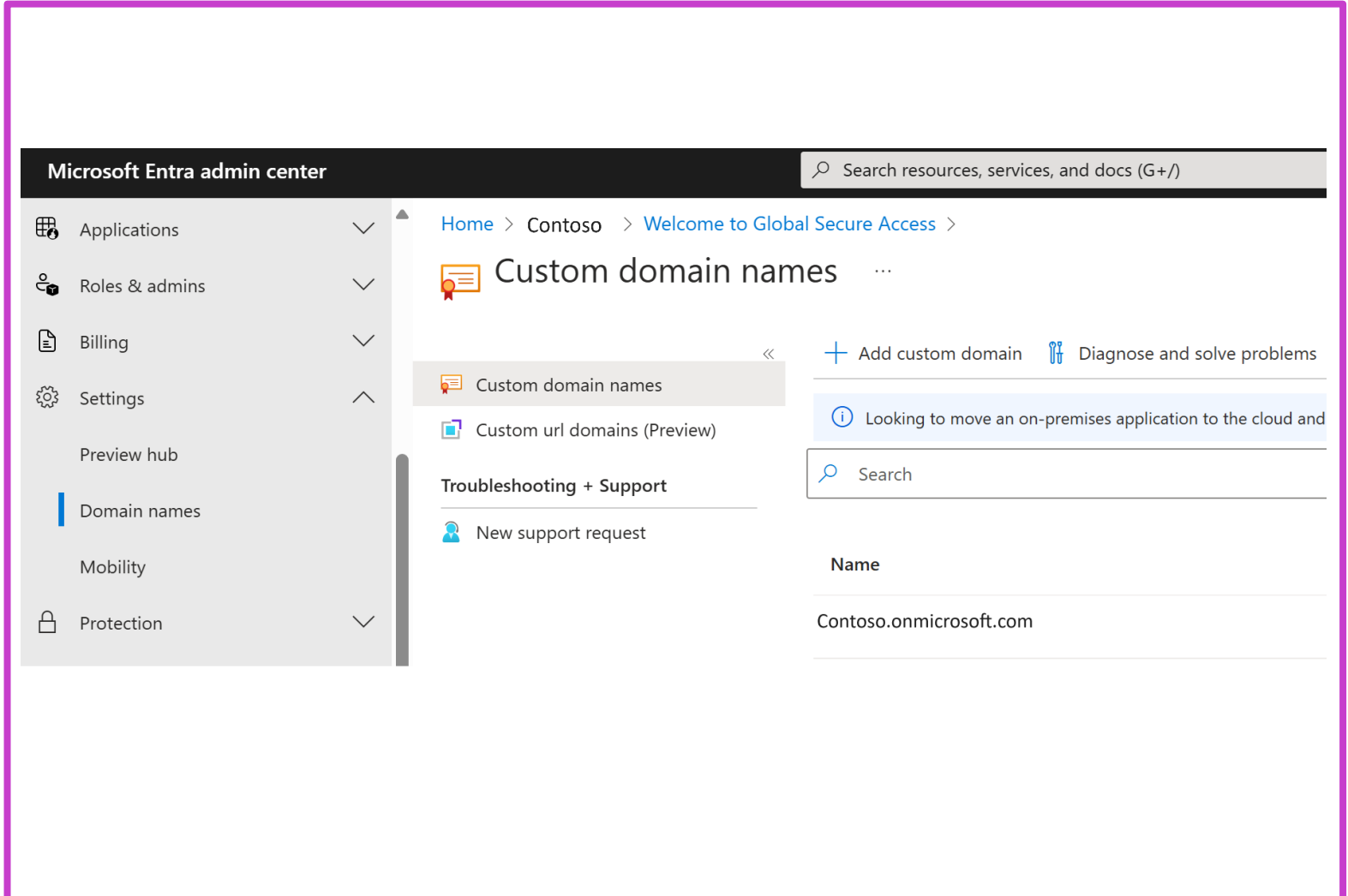
microsoft.azure.serviceHealth/allEntities/allTasks	Read and configure Azure Service Health
microsoft.azure.supportTickets/allEntities/allTasks	Create and manage Azure support tickets
microsoft.office365.copilot/allEntities/allProperties/allTasks	Create and manage all settings for Microsoft 365 Copilot
microsoft.office365.messageCenter/messages/read	Read messages in Message Center in the Microsoft 365 admin center, excluding security messages
microsoft.office365.network/performance/allProperties/read	Read all network performance properties in the Microsoft 365 admin center
microsoft.office365.search/content/manage	Create and delete content, and read and update all properties in Microsoft Search

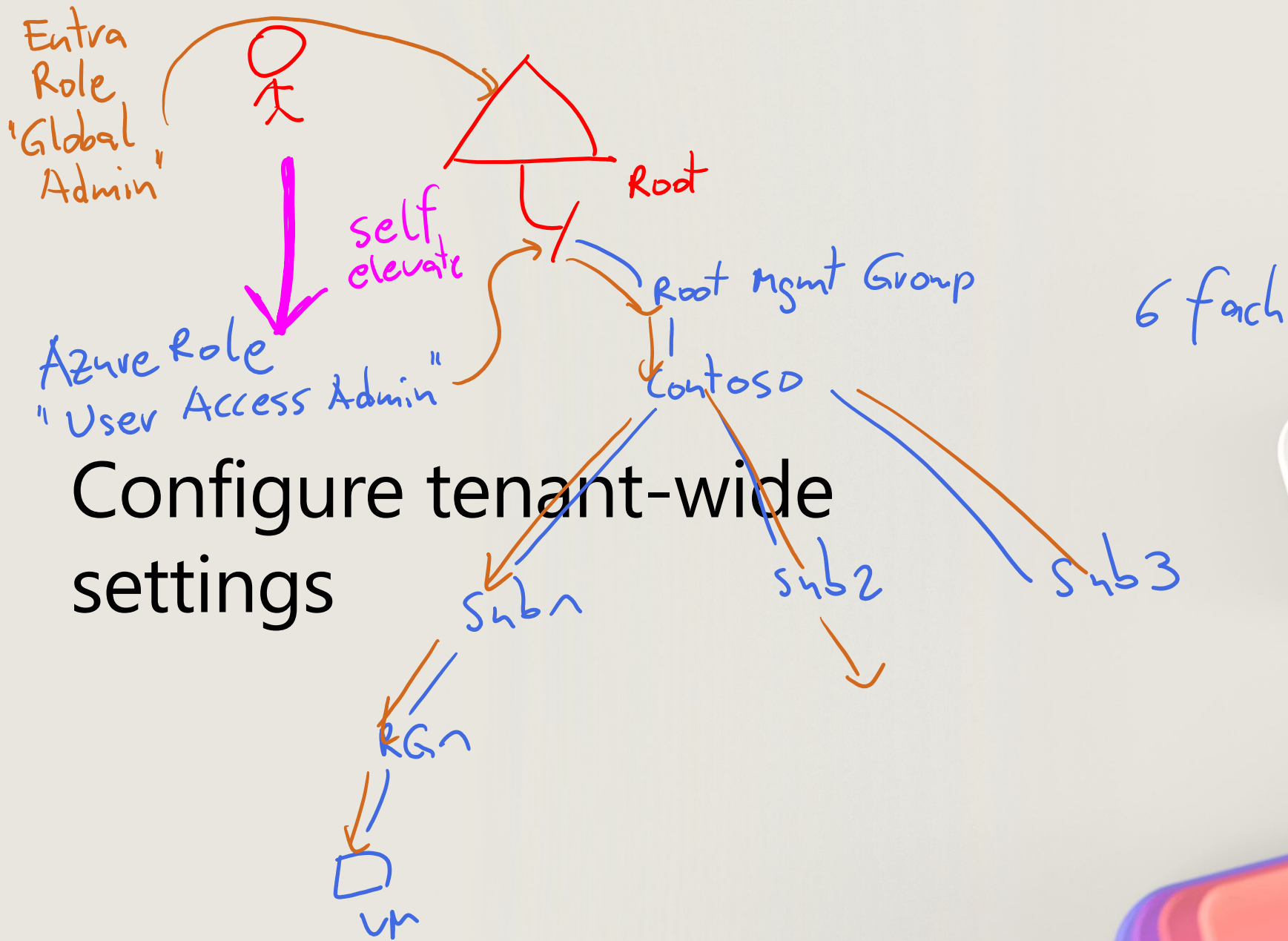
Configure and manage custom domains



Adding a custom domain name

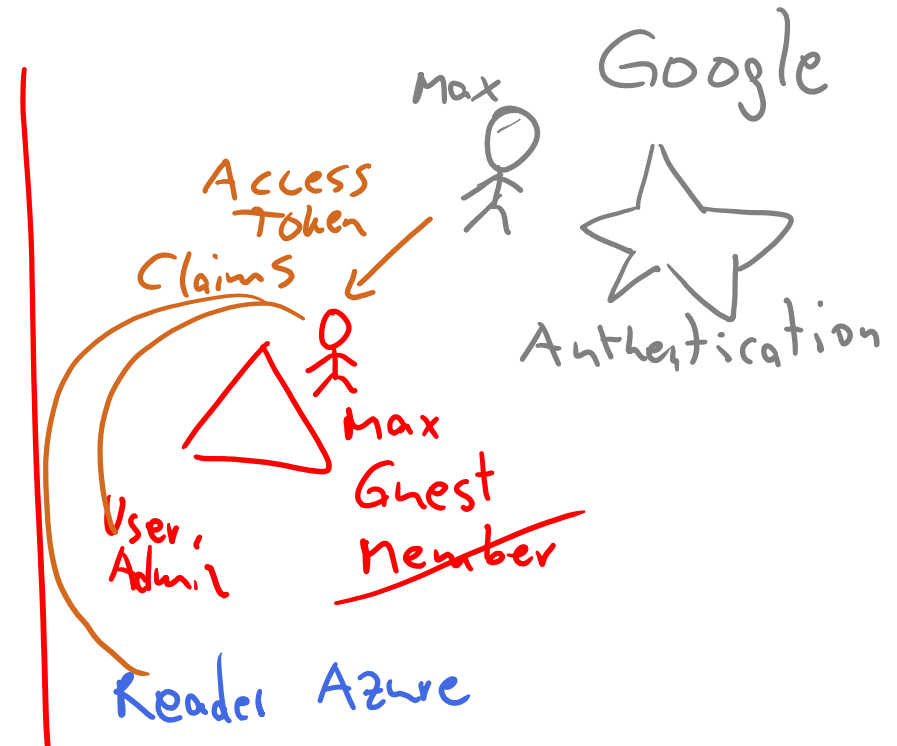
- Tenant has a default domain name ending in **onmicrosoft.com**
- Add your custom domain name and make it the primary domain name
- Add multiple custom domain names for an organization
- Add a root domain before adding subdomains





Tenant-wide settings

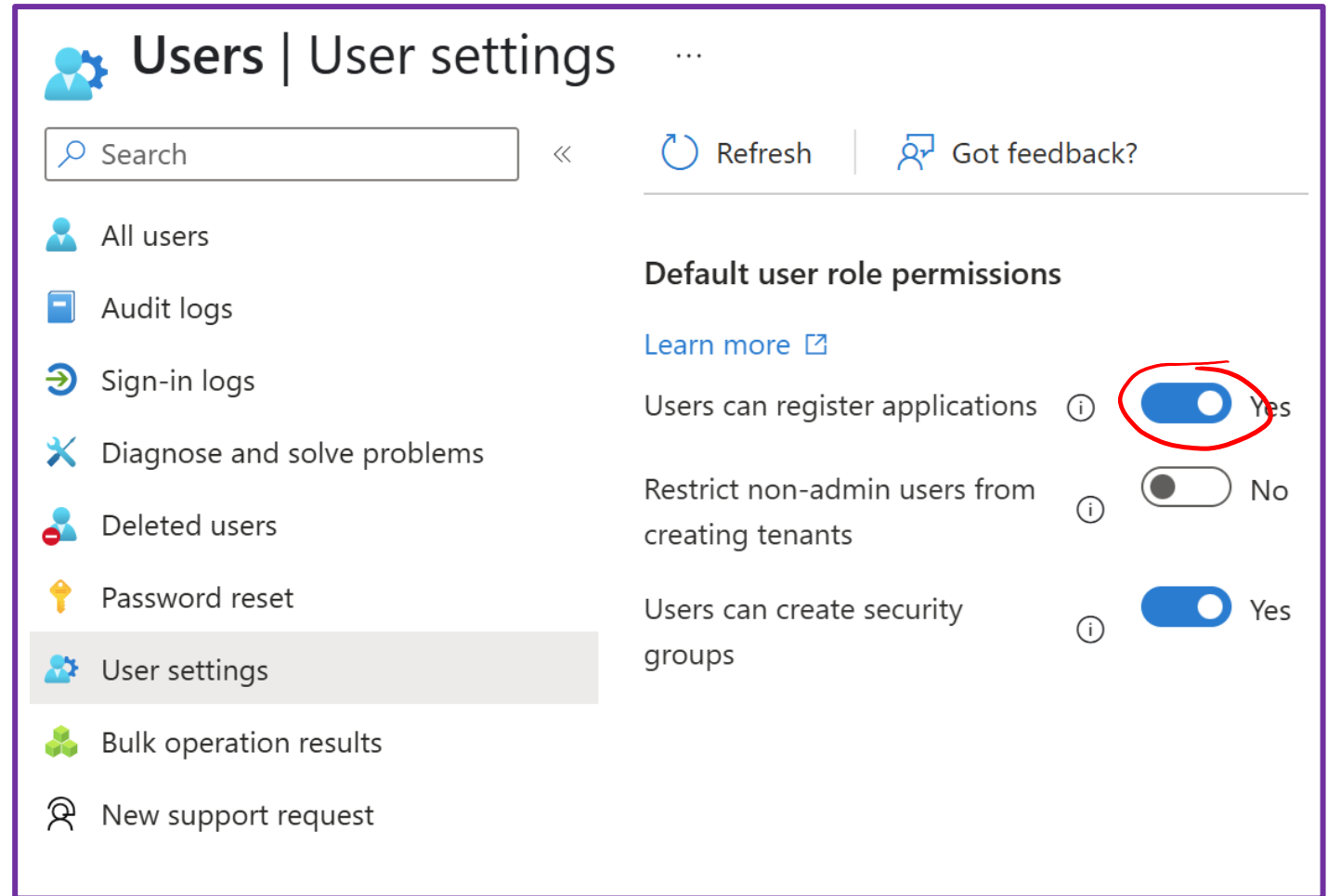
- Tenant-wide settings apply to the entire tenant when set
- These global values set some default behaviors to be applied to all users and properties
- Manage members and guest users:
 - Register applications
 - Restrict access to Microsoft Entra ID administrative portal
- Sign in with LinkedIn
- Manage security defaults *off* → *CA Policies*
- External collaboration *B2B* *OAuth 2.0*
 - Restricting who can invite guest and other settings



User settings

Microsoft Entra admin center →
Identity → Users → User Settings

- App registrations—users can register applications
- Administration portal – Restrict access to the Microsoft Entra ID administration portal
- LinkedIn account connections – Allow users to connect their work or school account with LinkedIn
- Other



User settings—external users

Microsoft Entra ID → Identity → Users → User Settings

- App registrations—users can register applications
- Administration portal—restrict access to the Microsoft Entra ID administration portal
- LinkedIn account connections—allow users to connect their work or school account with LinkedIn
- Manage user feature previews
- External collaboration setting

[Home](#) > [Default Directory](#) >

External collaboration settings ...

 Save  Discard

 Email one-time passcode for guests has been moved to All Identity Providers. →

Guest user access

Guest user access restrictions ⓘ

[Learn more](#)

- ☐ Guest users have the same access as members (most inclusive)
- ☐ Guest users have limited access to properties and memberships of directory objects
- ☒ Guest user access is restricted to properties and memberships of their own directory objects (most restrictive)

Guest invite settings

Guest invite restrictions ⓘ

[Learn more](#)

- ☐ Anyone in the organization can invite guest users including guests and non-admins (most inclusive)
- ☒ Member users and users assigned to specific admin roles can invite guest users including guests with member permissions
- ☐ Only users assigned to specific admin roles can invite guest users
- ☐ No one in the organization can invite guest users including admins (most restrictive)

Enable guest self-service sign up via user flows ⓘ

[Learn more](#)

Yes

No

Tenant properties

Entra Admin Center → Entra ID → Overview → Properties

- Changing the tenant display name
- Finding the country or region associated with your tenant
- Finding the location associated with your tenant
- Tenant ID
- Technical contact
- Global privacy contact
- Privacy statement URL

[Home](#) >

Default Directory

[+](#) Add [▼](#) [⚙️](#) Manage tenants [📄](#) What's new | [⚙️](#) Preview features

[i](#) Azure Active Directory is becoming Microsoft Entra ID. [Learn more](#) [🔗](#)

Overview Monitoring **Properties** Recommendations Tools

Name

Country or region

Data location

Notification language

Tenant ID

Technical contact

Global privacy contact

Privacy statement URL

Access management for Azure resources

Demo—set tenant-wide settings

Overview

Monitoring

Properties

Recommendations

Tutorials

Name

Contoso

Country or region

United States

Data location

United States datacenters

Notification language

English

Tenant ID

References (1 of 2)

Classic subscription administrator roles, Azure roles, and Microsoft Entra roles

<https://learn.microsoft.com/azure/role-based-access-control/rbac-and-directory-admin-roles>

Understand Azure role definitions

<https://learn.microsoft.com/azure/role-based-access-control/role-definitions>

Privileged Identity Management

<https://learn.microsoft.com/entra/id-governance/privileged-identity-management/>

How To: Configure the Microsoft Entra Multi-Factor Authentication registration policy

<https://learn.microsoft.com/entra/id-protection/howto-identity-protection-configure-mfa-policy>



References (2 of 2)

Conditional Access: Require MFA for administrators

<https://learn.microsoft.com/entra/identity/conditional-access/howto-conditional-access-policy-admin-mfa>

Conditional Access: Require MFA for Azure management

<https://learn.microsoft.com/entra/identity/conditional-access/howto-conditional-access-policy-azure-management>

Conditional Access: Block legacy authentication

<https://learn.microsoft.com/entra/identity/conditional-access/howto-conditional-access-policy-block-legacy>

Conditional Access: Require MFA for all users

<https://learn.microsoft.com/entra/identity/conditional-access/howto-conditional-access-policy-all-users-mfa>



Implement and manage external identities



Objectives

- 1 Describe guest users and B2B accounts
- 2 Manage external collaboration
- 3 How external users are managed in Microsoft 365
- 4 Invite external users—individually or in bulk
- 5 Manage external user accounts in Microsoft Entra ID
- 6 Implement cross-tenant access controls
- 7 Configure identity providers
- 8 Implement and manage Verified ID

Describe external users,
B2B accounts, and manage
external collaboration

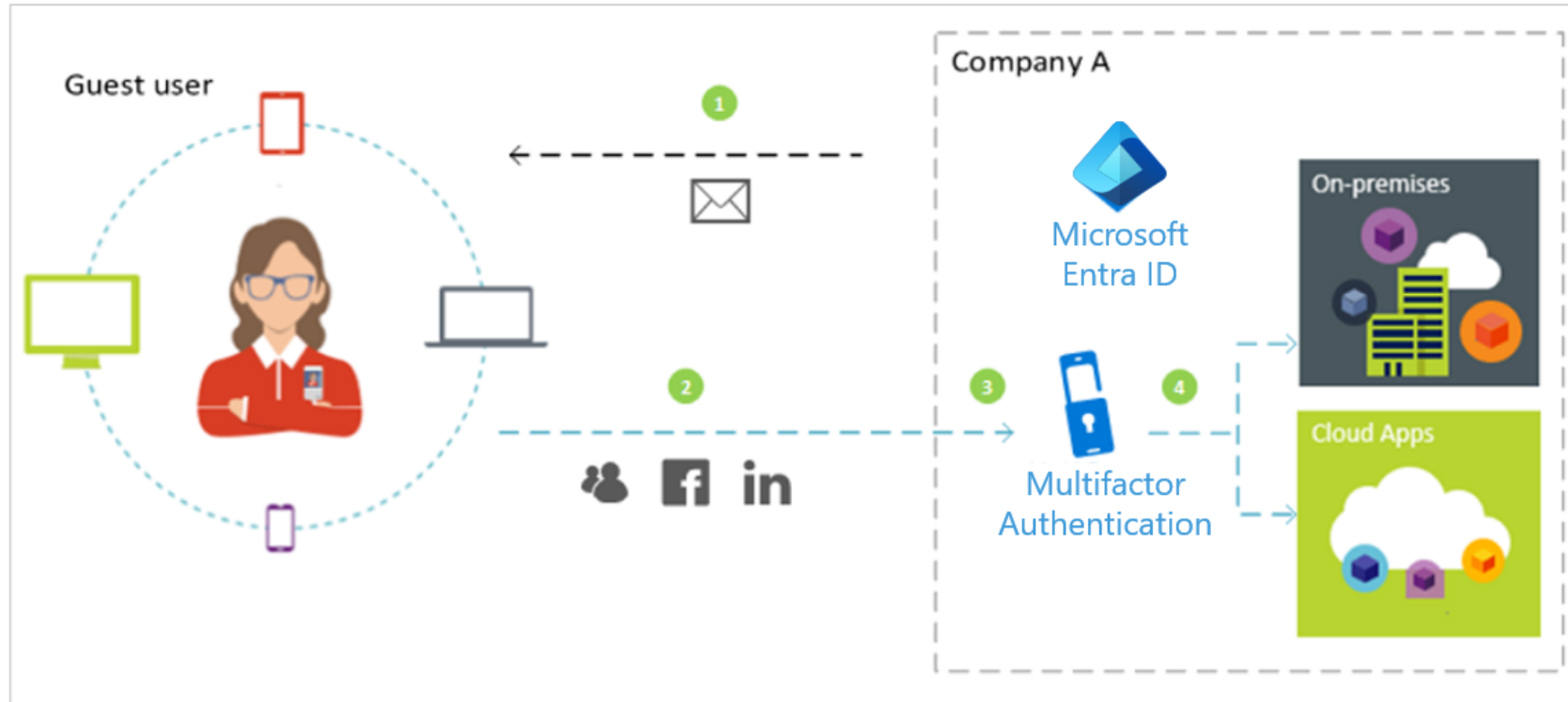


Relationship between external and member users

		UserType property	
		Guest	Member
How the user authenticates	External	External guest Uses an external Azure AD account, social identity, or other external identity provider to sign in. Most external users fall into this category.	External member Uses an external account to authenticate but has member-level access in your organization. Common scenario in multi-tenant organizations.
	Internal	Internal guest Has an account in your Azure AD directory but only guest-level access in your organization. This is often a legacy guest user created before the availability of Azure AD B2B.	Internal member Has an account in your Azure AD directory and member-level access in your organization. Generally considered employees of your organization.

Microsoft Entra B2B

B2C
App - Kunden DB



Guest user—a user invited to join your corporate Microsoft Entra ID.

Sourced from another directory, social media, partners, and other services.

Secure B2B collaboration projects enabled.

Collaborating with external users

- Invite external users into your tenant typically as guests
- External users use their existing credentials for authentication
- They are assigned permissions for authorization
- You can restrict what external users can see and do

Exercise—configure external collaboration



This exercise teaches students how to configure external collaboration settings.

[Launch this Exercise in GitHub](#)

Guest invite settings

Guest invite restrictions ⓘ

[Learn more](#)

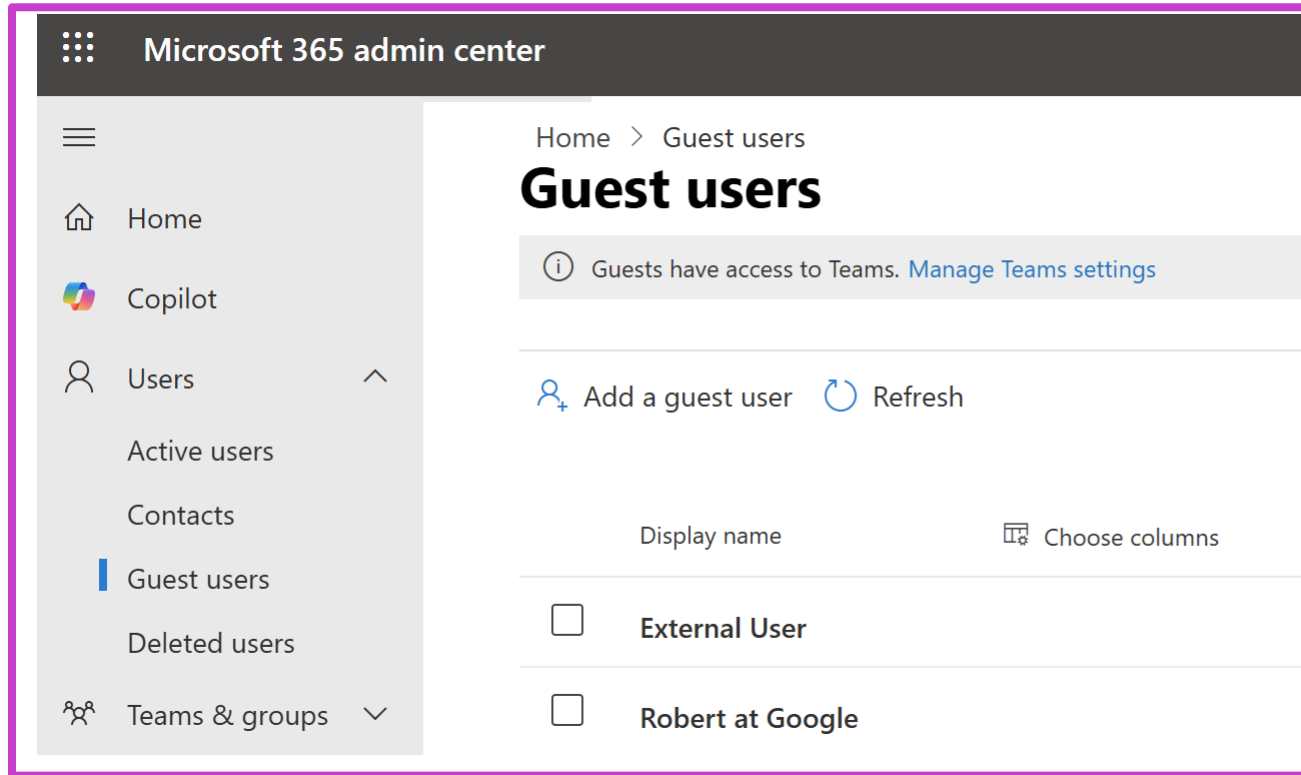
- ☒ Anyone in the organization can invite guest users including guests and non-admins (most inclusive)
- ☐ Member users and users assigned to specific admin roles can invite guest users including guests with member permissions
- ☐ Only users assigned to specific admin roles can invite guest users
- ☐ No one in the organization can invite guest users including admins (most restrictive)

Enable guest self-service sign up via user flows ⓘ

External users from M365 workloads



Microsoft 365 admin center



Manage guest users in the Microsoft 365 admin center.

Similar process to Microsoft Entra ID.

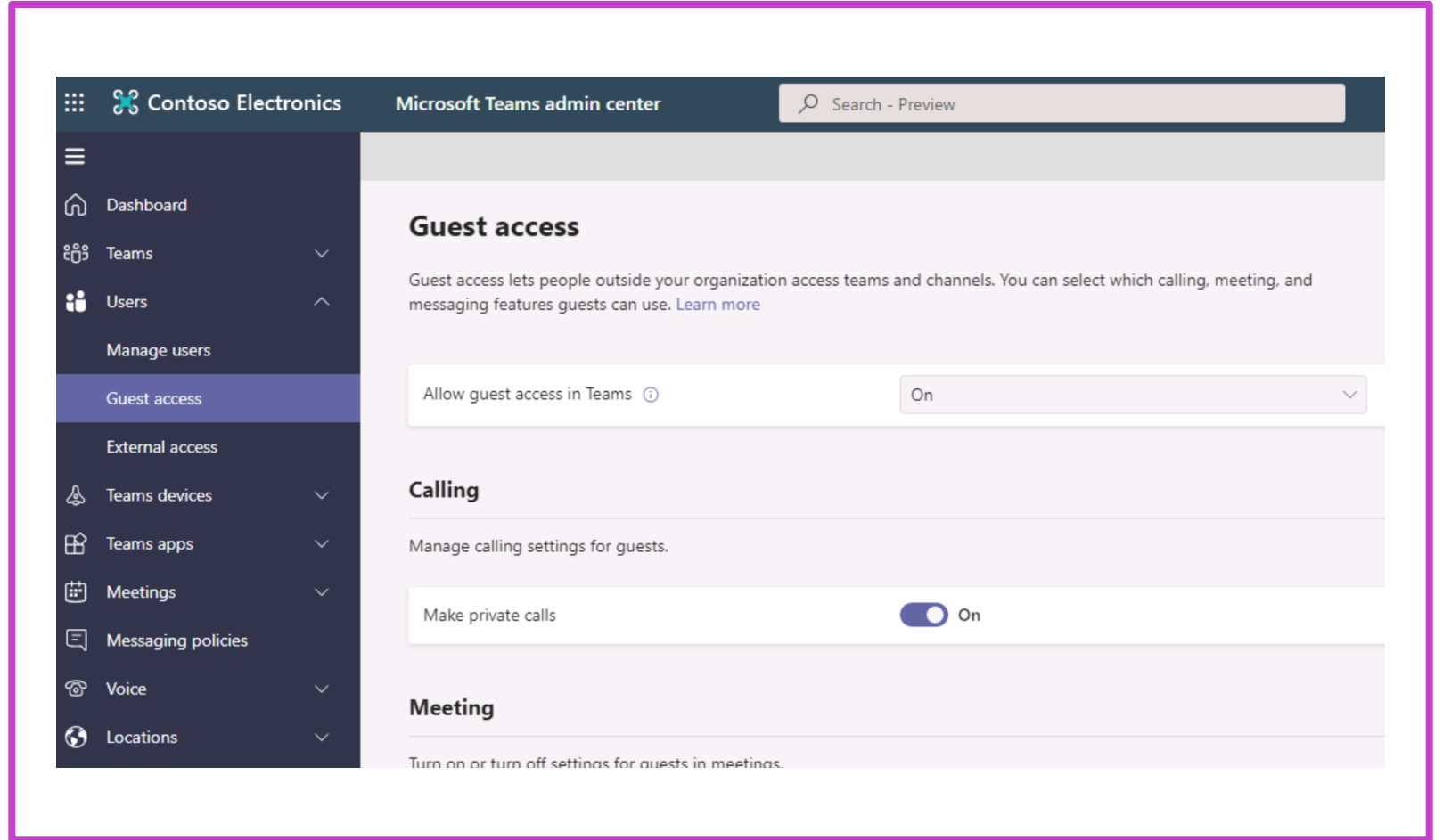
Always use Zero Trust principles when managing.

External collaboration options in Microsoft 365

Activity	Account type	Default setting
Authenticated file and folder sharing	Guest account	Enabled
Site sharing	Guest account	Enabled
Team sharing	Guest account	Enabled
Shared channel in Teams	Existing Microsoft 365 external account	Disabled
External chat and meetings	Existing Microsoft 365 external account	Enabled
Anonymous meeting join	None	Enabled
Unauthenticated file and folder sharing	None	Enabled

Granting/removing external user access

- Configure at the workload admin level
- Always have a governance process in place
- Always use Zero Trust principles when managing



Invite external users—
Individually or in bulk



Inviting users

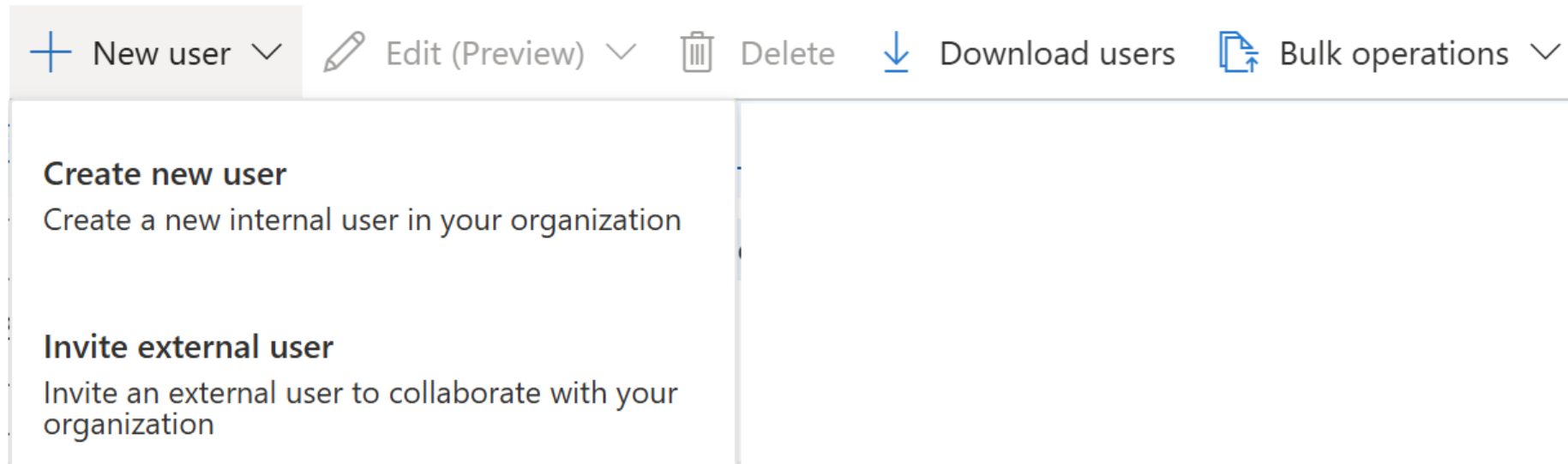
Inviting users individually:

- Any user (even a guest) can invite guest users by default
- Inviter sends the guest a direct link to the app being shared
- Application owners can manage their own guest users

Inviting users in bulk:

- Prepare a.csv file with user information and invitation preferences
- Upload the.csv file to Microsoft Entra ID

Demo—add external user (single or bulk)



Manage external user accounts in Microsoft Entra ID

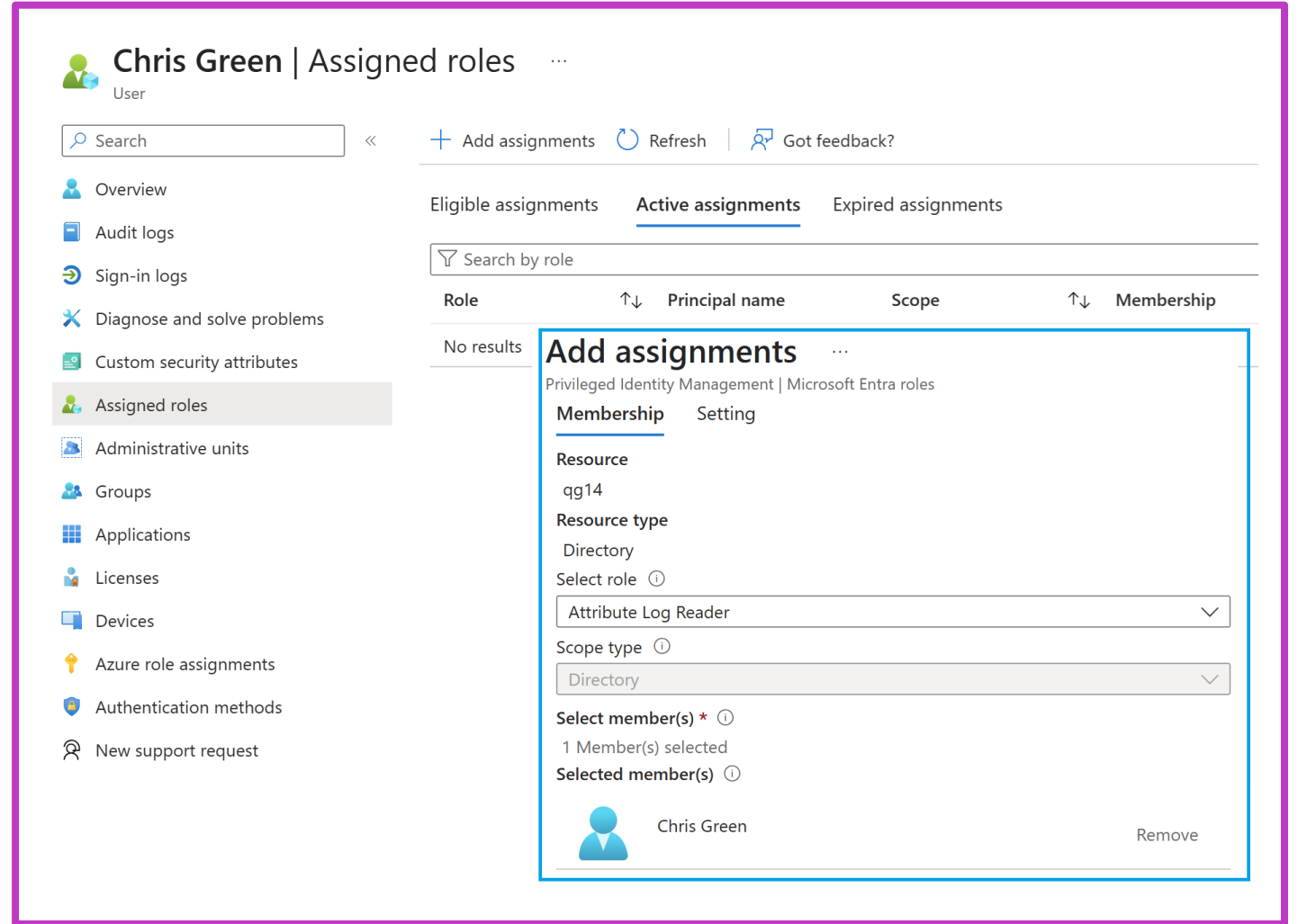


External users—restricted by default

External users do not have:

- Assigned roles
- Group membership
- Licenses

Grant access as your security posture and organization needs dictate using Entitlement or Manually.



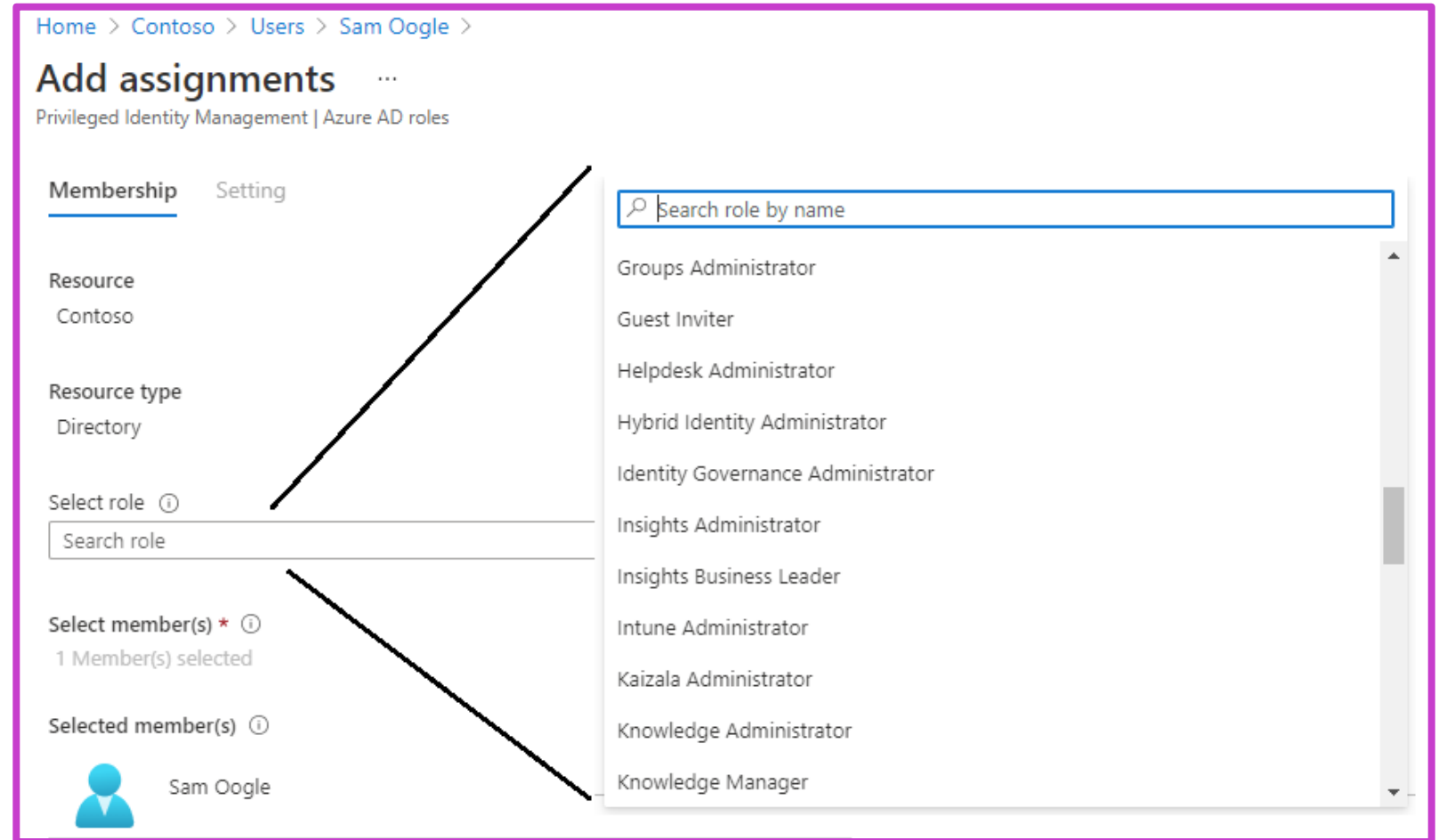
The screenshot shows the Microsoft Entra Admin Center interface for user Chris Green. The page is titled "Chris Green | Assigned roles". The left sidebar contains a navigation menu with options: Overview, Audit logs, Sign-in logs, Diagnose and solve problems, Custom security attributes, Assigned roles (selected), Administrative units, Groups, Applications, Licenses, Devices, Azure role assignments, Authentication methods, and New support request. The main content area shows the "Assigned roles" page with a search bar and buttons for "Add assignments", "Refresh", and "Got feedback?". The page is divided into three tabs: "Eligible assignments", "Active assignments" (selected), and "Expired assignments". Below the tabs is a search bar labeled "Search by role". A table with columns "Role", "Principal name", "Scope", and "Membership" is shown, but it contains "No results". A modal titled "Add assignments" is open, showing the "Membership" tab. The modal displays the resource "qg14" and the resource type "Directory". The "Select role" dropdown is set to "Attribute Log Reader". The "Scope type" dropdown is set to "Directory". The "Select member(s)" section shows "1 Member(s) selected" and lists "Chris Green" with a "Remove" button.

Provide the guest user the least privilege access needed

Add guest to:

- "Assign roles" just as with normal users.

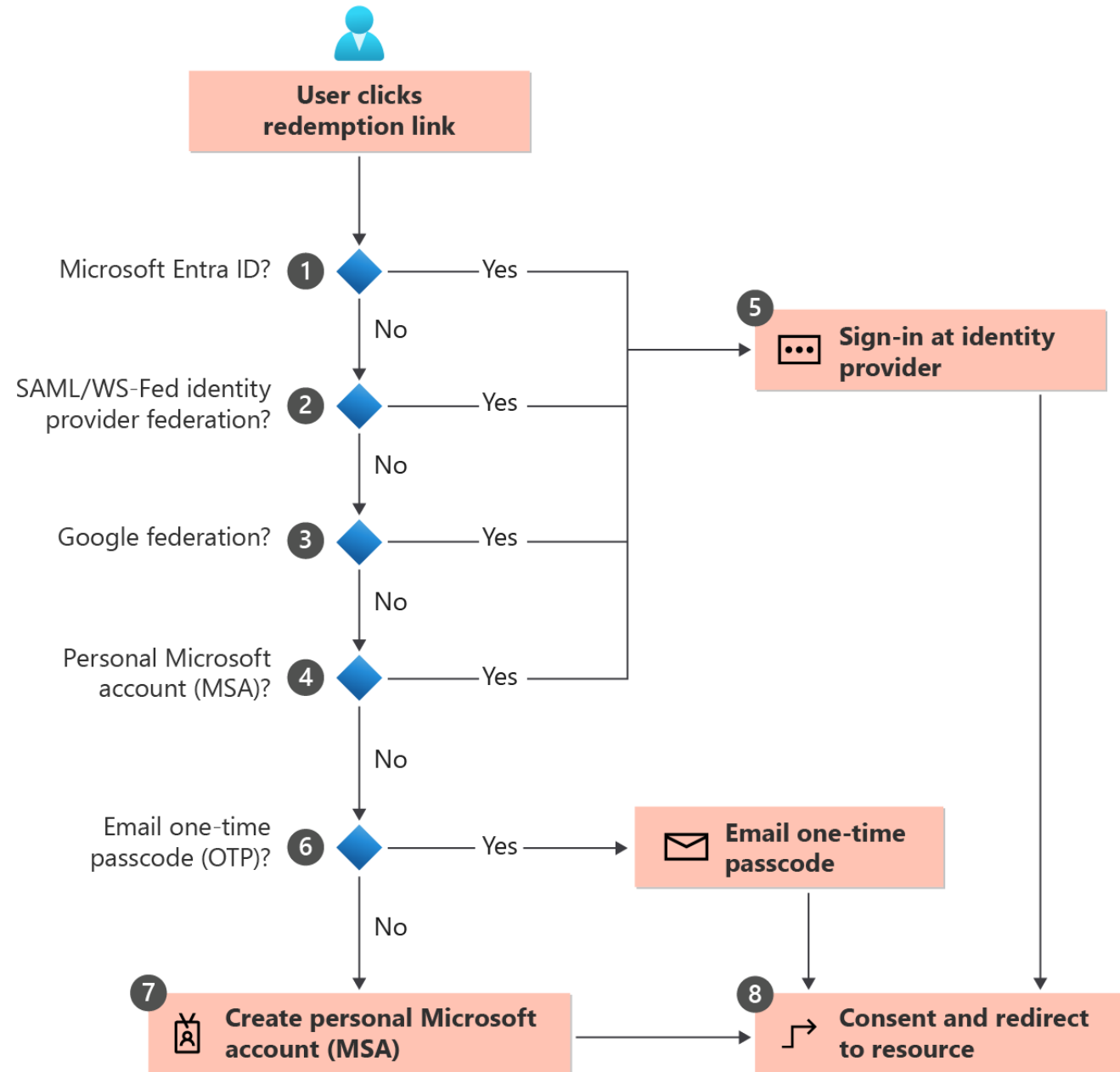
You can add Guests to groups as well.



B2B user redemption

External identity remediation types

- Microsoft Entra ID native
- SAML / WS-Fed
- Google (other social media)
- Personal MSA
- One-time passcode



Cross-tenant access controls



Configure cross-tenant access controls

- Default—applied to all external collaboration.
- Organizational settings—add rules by specific tenant/tenant-id.
- Microsoft cloud—control access to Azure Government clouds.

Home > External Identities

External Identities | Cross-tenant access settings

Search << Got feedback?

Overview

Cross-tenant access settings

All identity providers

External collaboration settings

Diagnose and solve problems

Self-service sign up

Custom user attributes

All API connectors

Custom authentication extensions (Preview)

User flows

Subscriptions

Linked subscriptions

Lifecycle management

Terms of use

Access reviews

Troubleshooting + Support

Organizational settings

Default settings

Microsoft cloud settings

Default settings apply to all external Azure AD organizations not listed on the organizational settings tab. These default settings [Learn more](#)

Inbound access settings

Edit inbound defaults

Type	Applies to	Status
B2B collaboration	External users and groups	All allowed
B2B collaboration	Applications	All allowed
B2B direct connect	External users and groups	All blocked
B2B direct connect	Applications	All blocked
Trust settings	N/A	Disabled

Outbound access settings

Edit outbound defaults

Configure cross-tenant synchronization

Synchronize users and attributes across organizations

- Automatically create and delete users cross-tenant.
- Move attributes and settings across tenants.
- Synced users benefit from features like Conditional Access and analytics.
- Push only—from source tenant to a destination tenant.

[Home](#) > [Default 2 | Cross-tenant synchronization](#) > [Cross-tenant synchronization | Configurations](#) > [Fabrikam | Overview](#) >

Provisioning

 Save  Discard

Provisioning Mode

Automatic

Use Azure AD to manage the creation and synchronization of user accounts in Fabrikam based on user and group assignment.

Admin Credentials

Admin Credentials

Azure AD needs the following information to connect to Fabrikam's API and synchronize user data.

Authentication Method ⓘ

Cross Tenant Synchronization Policy

Tenant Id *

Test Connection

Configure identity providers



SAML/WS-Fed

- You can set up direct federation with any organization that supports SAML 2.0 or WS-Fed
- Any new guests will be authenticated using direct federation
- External users sign in using their own organizational account

SAML/WS-Fed identity provider configuration

Home > Users > External Identities

External Identities | All identity providers

Search

Overview

Cross-tenant access settings

All identity providers

External collaboration settings

Diagnose and solve problems

Self-service sign up

Custom user attributes

All API connectors

Custom authentication extensions

User flows

Built-in Custom

Name	Status
Microsoft Entra ID	Configured
Email one-time passcode	Configured
Microsoft	Configured
Google	Configured
Facebook	Configure

New SAML/WS-Fed IdP

You must configure the federating identity provider first. →

Display name *

Identity provider protocol *

Domain name of federating IdP *

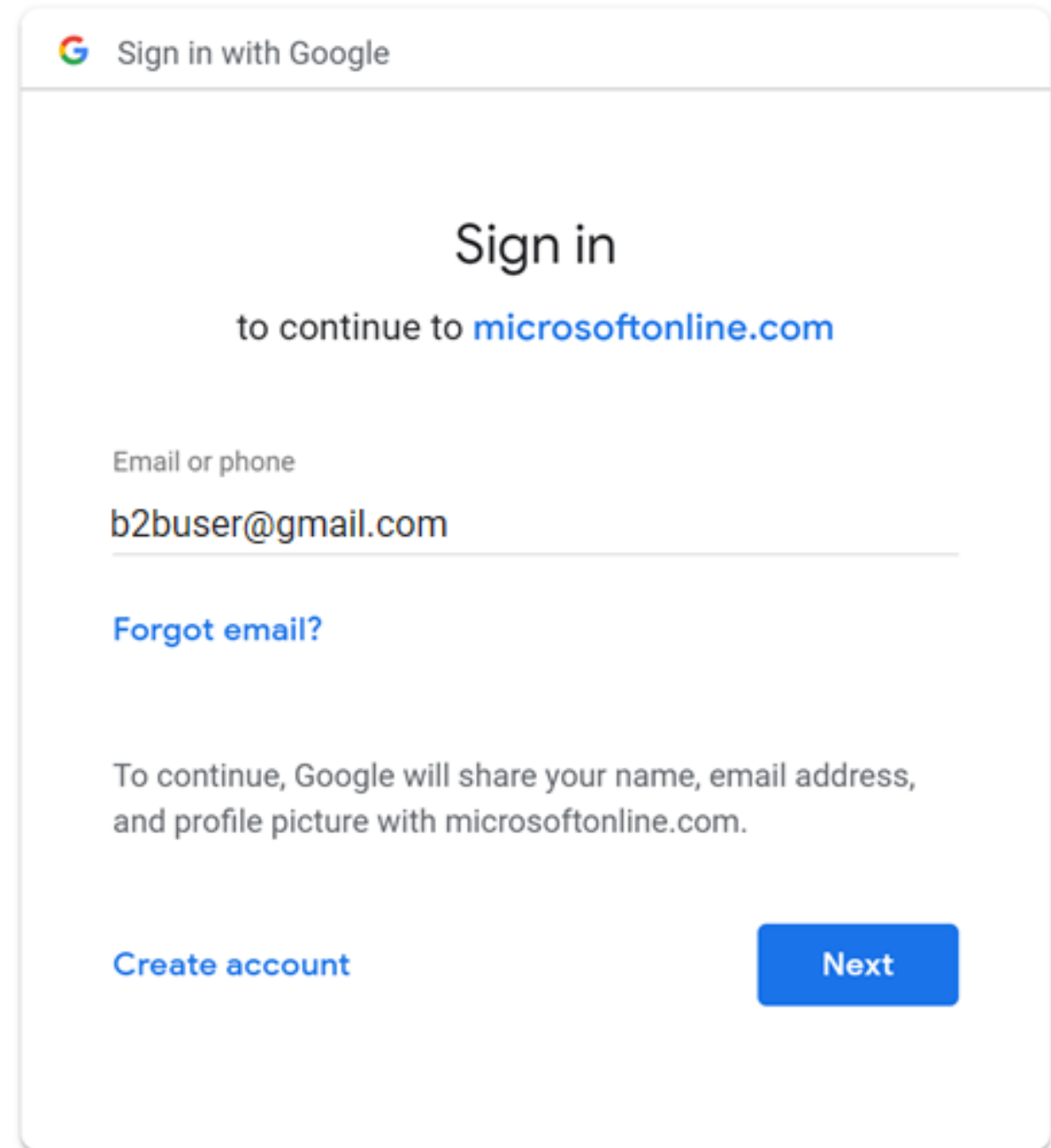
You can add more domains after configuring the SAML/WS-Fed IdP. [Learn more.](#)

Select a method for populating metadata *

Save Cancel

Using Google as an identity provider

- External users who aren't signed in to Google will be prompted to do so
- External users who are already signed in to Google will be prompted to choose the account they want to use
- They must choose the account you used to invite them



The image shows a Google sign-in interface. At the top, there is a header with the Google logo and the text "Sign in with Google". Below this, the main heading is "Sign in" followed by "to continue to [microsoftonline.com](#)". There is a text input field labeled "Email or phone" containing the email address "b2buser@gmail.com". Below the input field, there is a link "Forgot email?". At the bottom, there is a link "Create account" and a blue button labeled "Next".

Sign in with Google

Sign in

to continue to [microsoftonline.com](#)

Email or phone

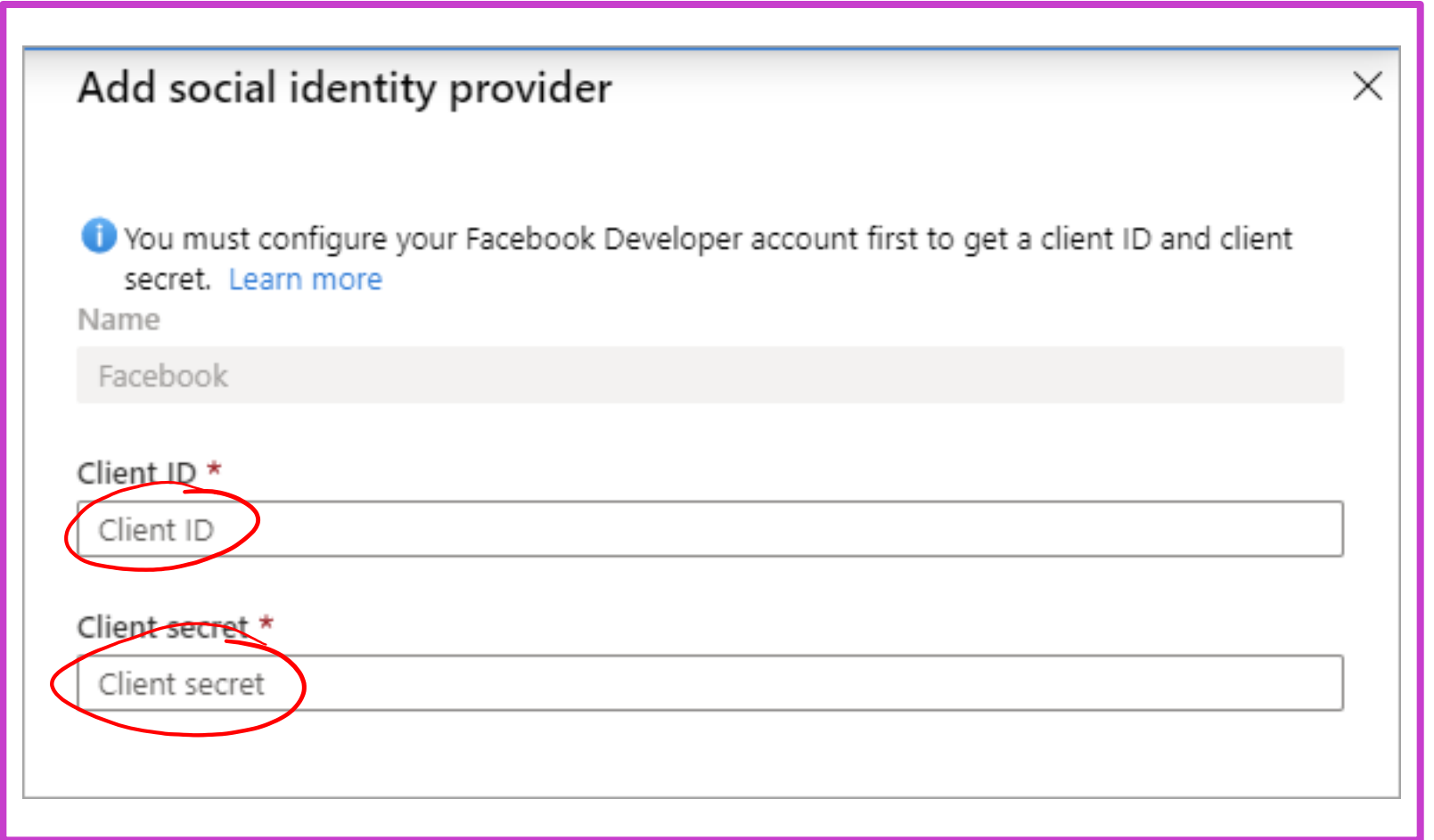
[Forgot email?](#)

To continue, Google will share your name, email address, and profile picture with microsoftonline.com.

[Create account](#) [Next](#)

Using Facebook as an identity provider

- To use a Facebook account as an identity provider, you must create an application in the Facebook developers' console
- Set the Facebook client ID and client secret using either the Microsoft Entra ID portal or PowerShell
- You can test your Facebook configuration by signing up via a user flow on an app enabled for self-service sign-up



Add social identity provider [X]

i You must configure your Facebook Developer account first to get a client ID and client secret. [Learn more](#)

Name

Facebook

Client ID *

Client ID

Client secret *

Client secret



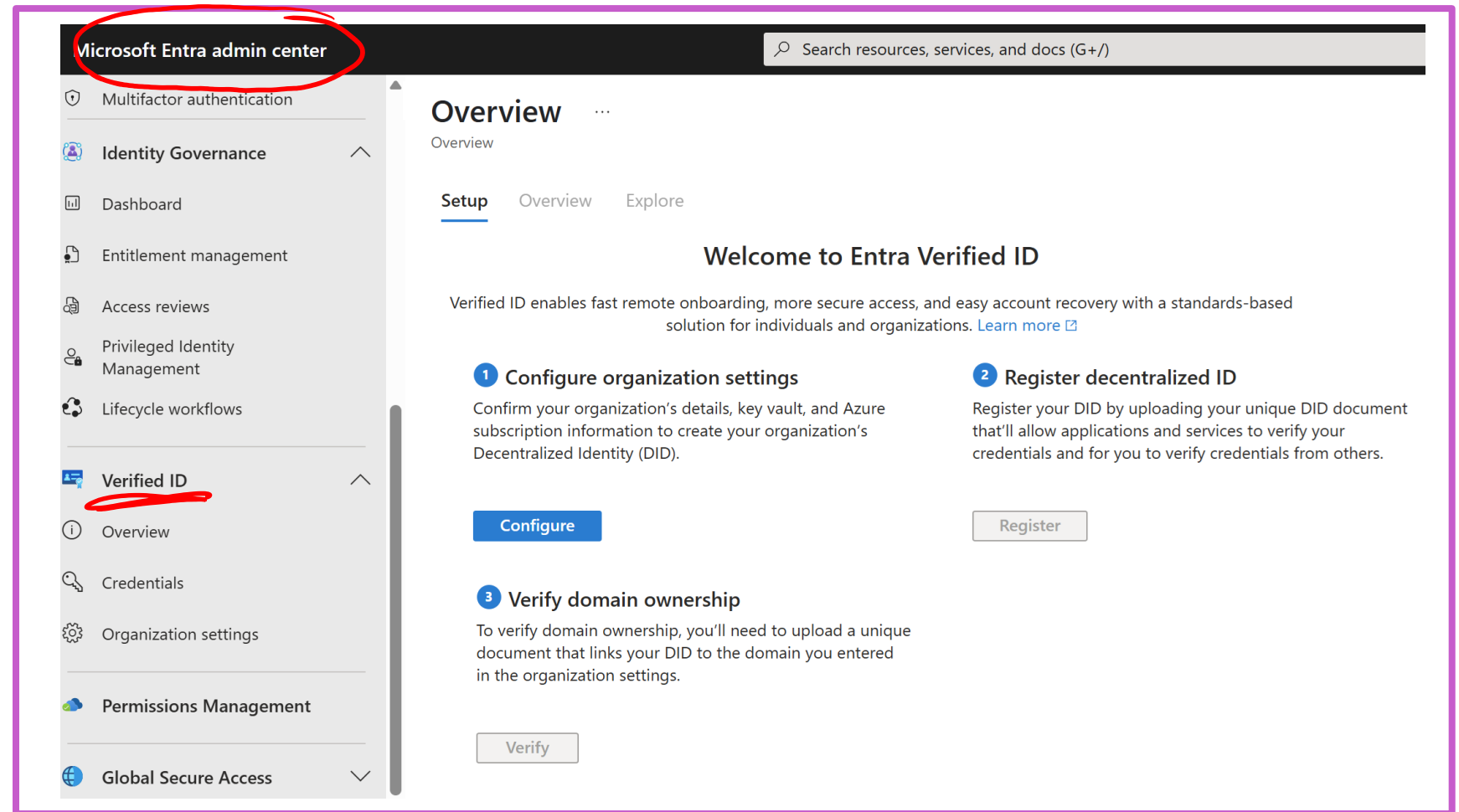
Verifiable credentials



Implement Microsoft Entra Verified ID

Requirements:

- Azure subscription
- Microsoft Entra ID Premium
- Azure Key Vault



Microsoft Entra Verified ID usage and value

Value

- **Trust without over sharing**
 - Share only the information needed for verification
- **Decentralized / tamper resistant**
 - Cryptographically signed and held by the users (not company)
- **Portable Trust**
 - Usable across systems and platforms
- **Extends authentication and Authorization**
 - Compliments Entra ID existing verifications

Usage Scenarios

- **Employee verification**
 - Temp staff, contractors, shared desks
 - Blocks badge cloning / theft
- **B2B Guest support**
 - No longer need long lived guest accounts
- **On boarding**
 - New hires
 - Allow new hires to start with biometrics enabled
- **Secure API and AI access**
 - Require biometrics for additional access

:88 Kerberos

:443

○ Auth 2.0
Entra ID

sync

write back

PHS*

Password Hash Sync.

Windows
Server

Entra Connect

On Prem

~~password 123~~

SHA256
+ RC4

Implement and manage hybrid identity

Connect Sync.

PTA

WS-Fed

New: Cloud Sync.
No Devices

Objectives

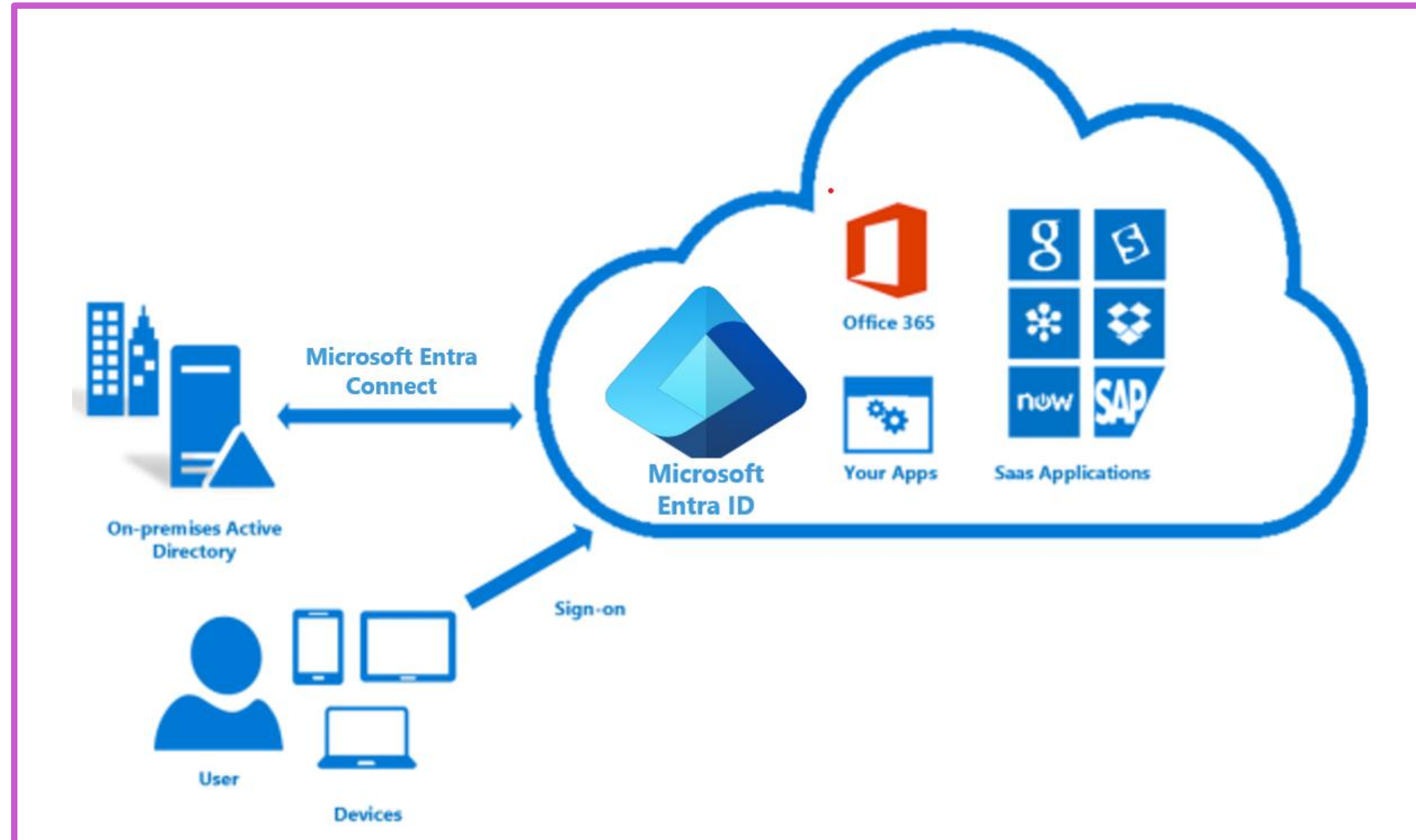
- 1 Plan, design, and implement Microsoft Entra Connect Sync
- 2 Implement and manage password hash synchronization (PHS)
- 3 Implement and manage pass-through authentication (PTA)
- 4 Implement and manage federation
- 5 Troubleshoot synchronization errors
- 6 Implement Microsoft Entra Connect Health
- 7 Manage Microsoft Entra Connect Health

Plan, design, and implement Microsoft Entra Connect



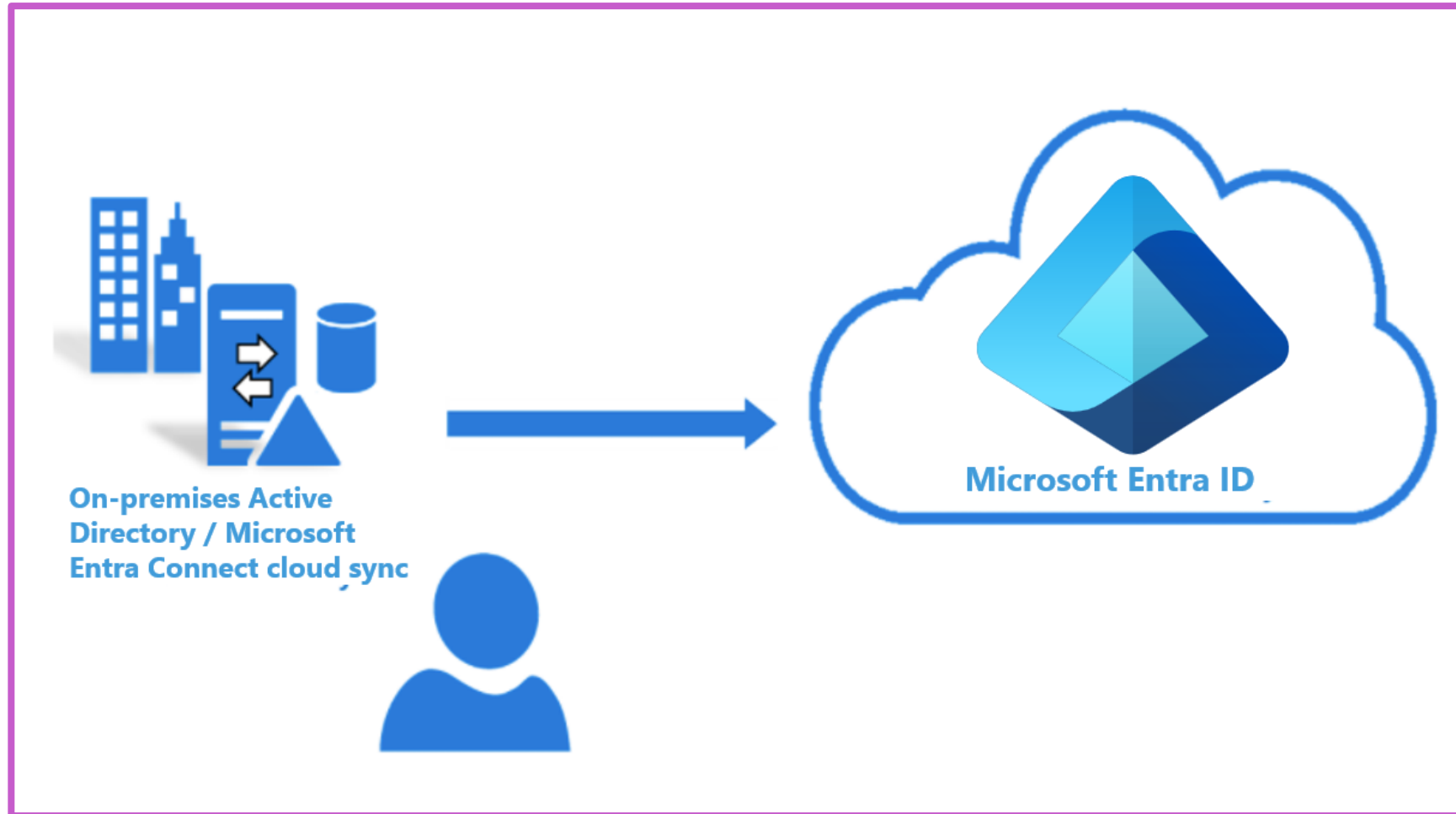
What is Microsoft Entra Connect?

- Microsoft Entra Connect is a solution that bridges an organization's on-premises Active Directory with your cloud-based Microsoft Entra ID
- Microsoft Entra Connect provides:
 - Synchronization
 - Password hash synchronization
 - Pass-through authentication
 - Federation integration
 - Health monitoring



Microsoft Entra cloud sync

- Microsoft Entra cloud sync is a solution that syncs your on-premises AD with Microsoft Entra ID
- Lightweight provisioning agent required on the on-premises AD
- All sync configuration is managed in the cloud
- Can be used in conjunction with Microsoft Entra Connect



Authentication methods

Cloud authentication

Microsoft Entra password hash synchronization (PHS)

- Users can use the same username and password that they use on-premises

Microsoft Entra pass-through authentication (PTA)

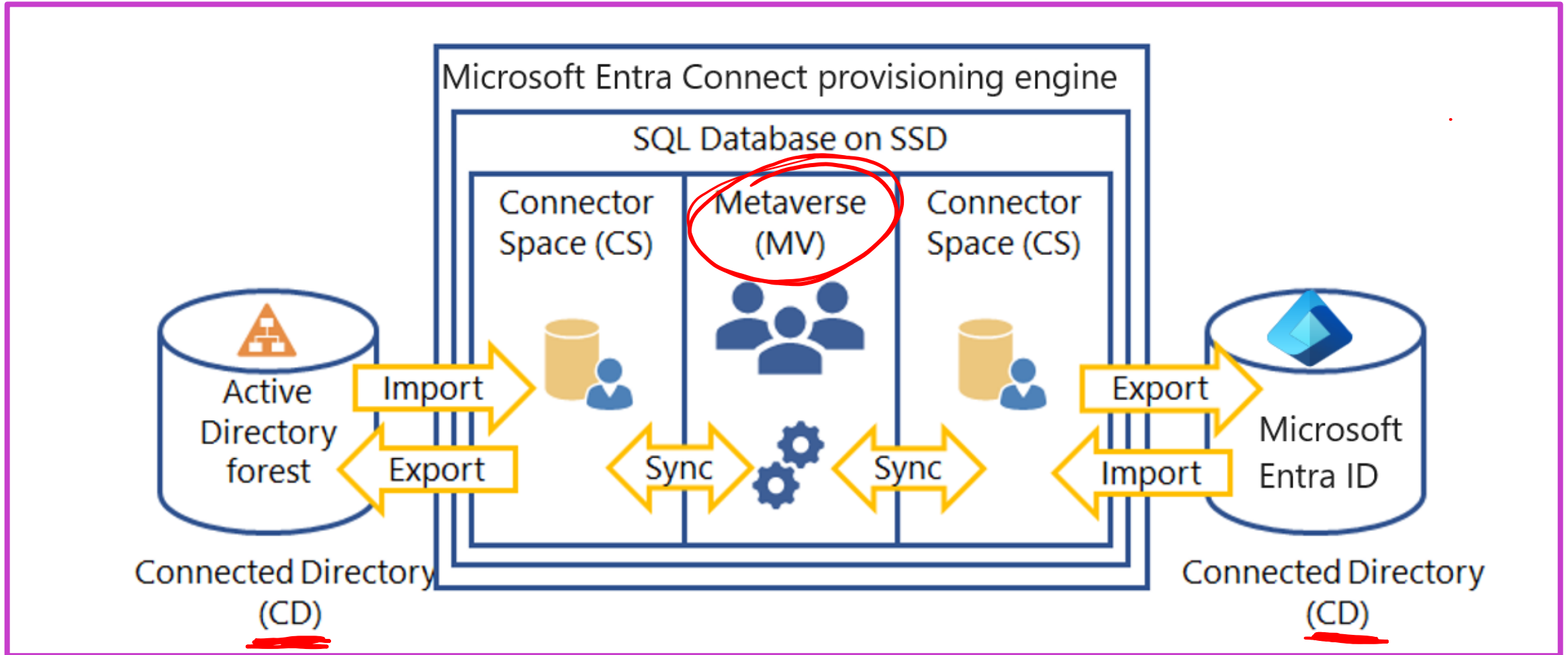
- Simple password validation for Microsoft Entra ID authentication services using a software agent that runs on one or more on-premises servers

Federated authentication

Handoff to trusted authentication system

- Microsoft Entra ID hands off the authentication process to a separate trusted authentication system to validate the user's password
- The authentication system can provide additional advanced authentication requirements, such as a smartcard or third-party multifactor authentication

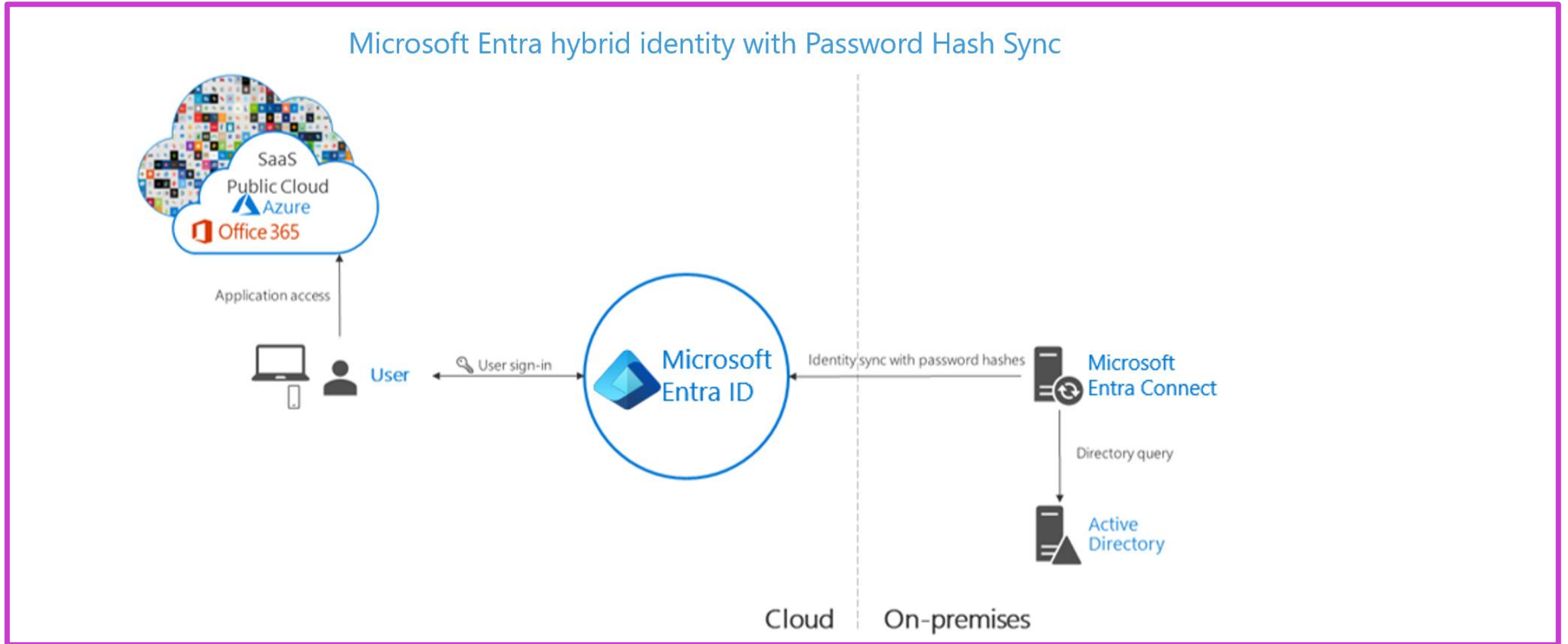
Microsoft Entra Connect component factors



Implement and manage password hash synchronization (PHS)

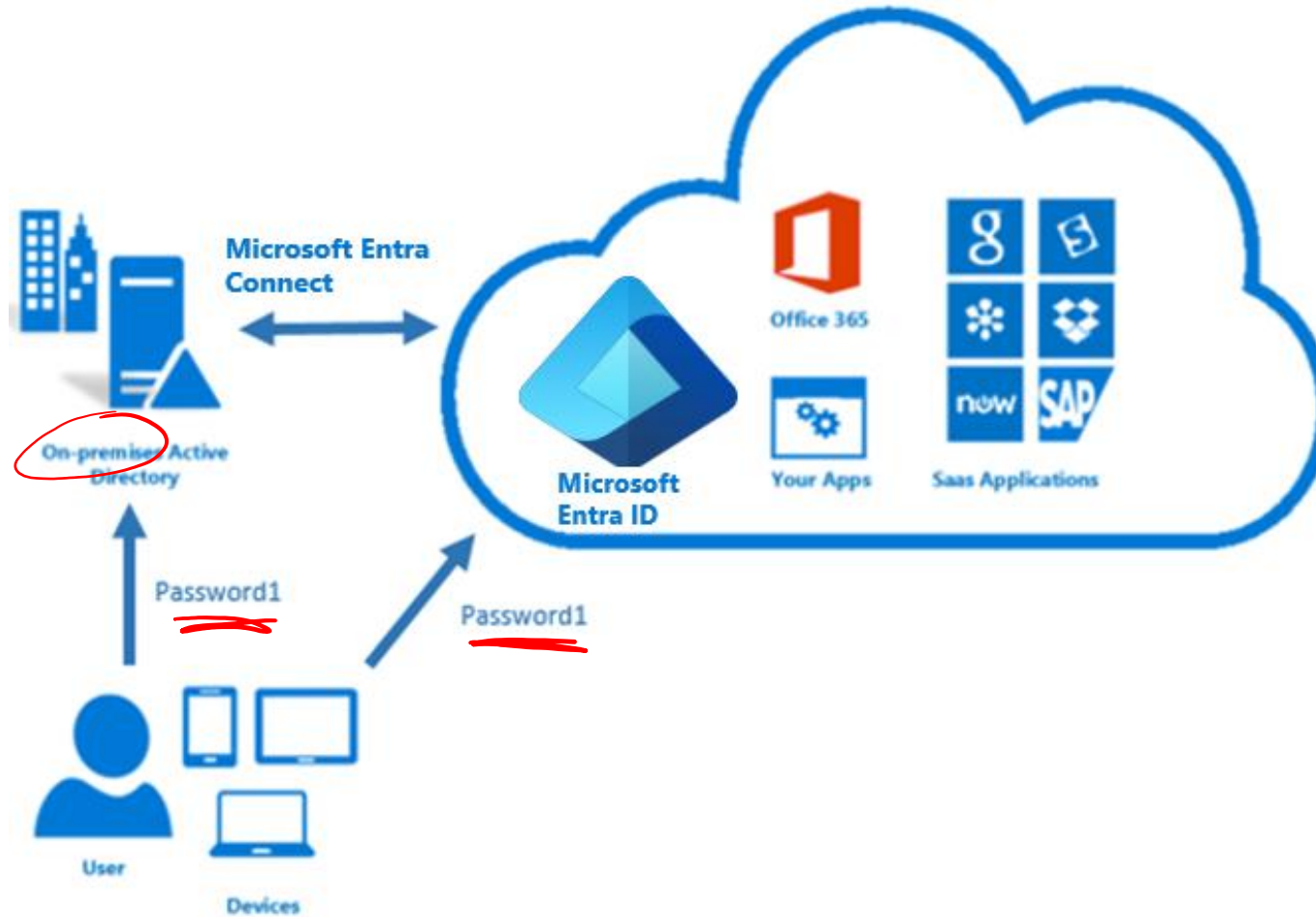


Password hash sync

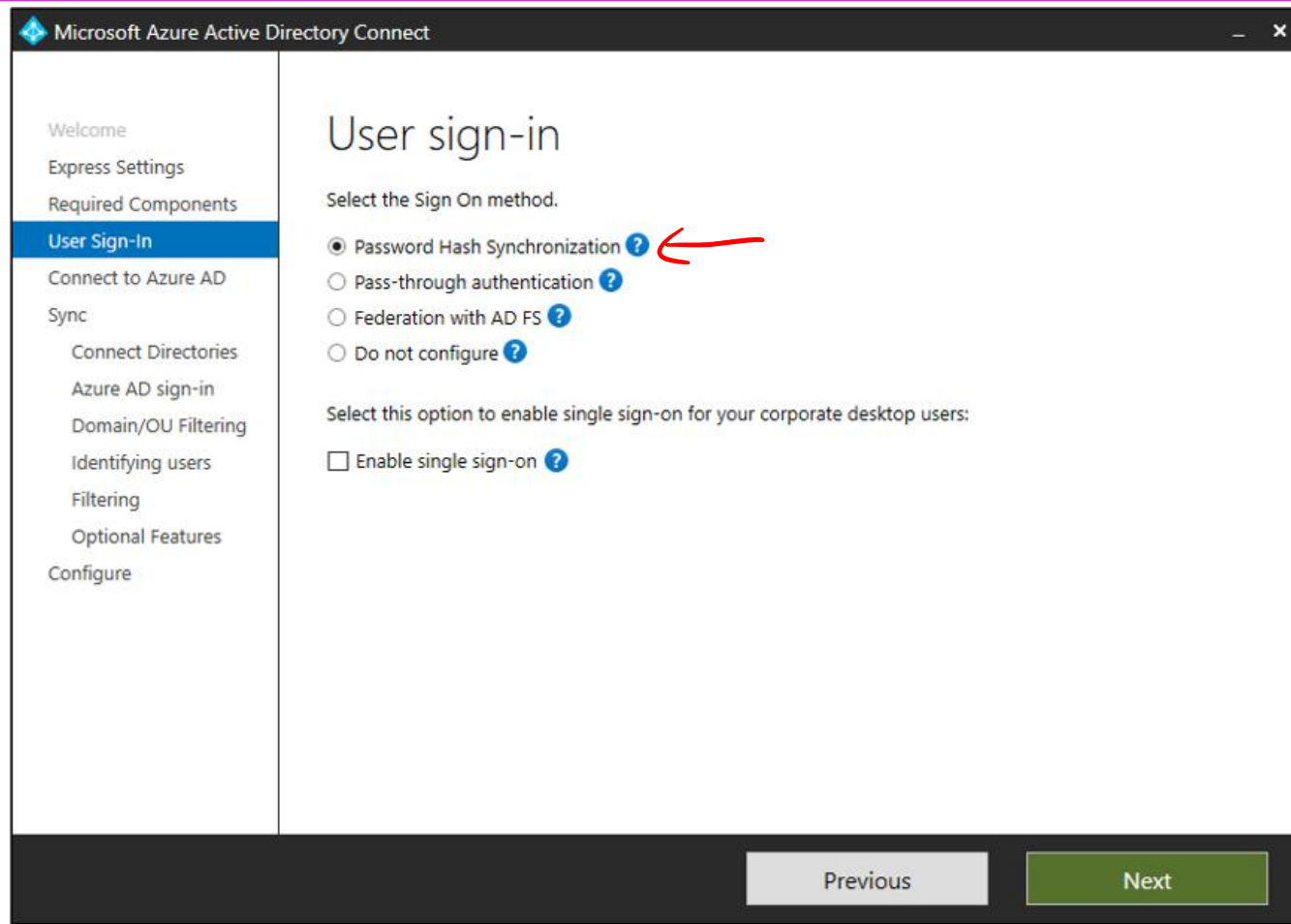


Simplicity of a password hash synchronization solution

How password hash synchronization works



Enable password hash synchronization



Microsoft Azure Active Directory Connect

Welcome
Express Settings
Required Components
User Sign-in
Connect to Azure AD
Sync
Connect Directories
Azure AD sign-in
Domain/OU Filtering
Identifying users
Filtering
Optional Features
Configure

User sign-in

Select the Sign On method.

- ☒ Password Hash Synchronization ?
- ☐ Pass-through authentication ?
- ☐ Federation with AD FS ?
- ☐ Do not configure ?

Select this option to enable single sign-on for your corporate desktop users:

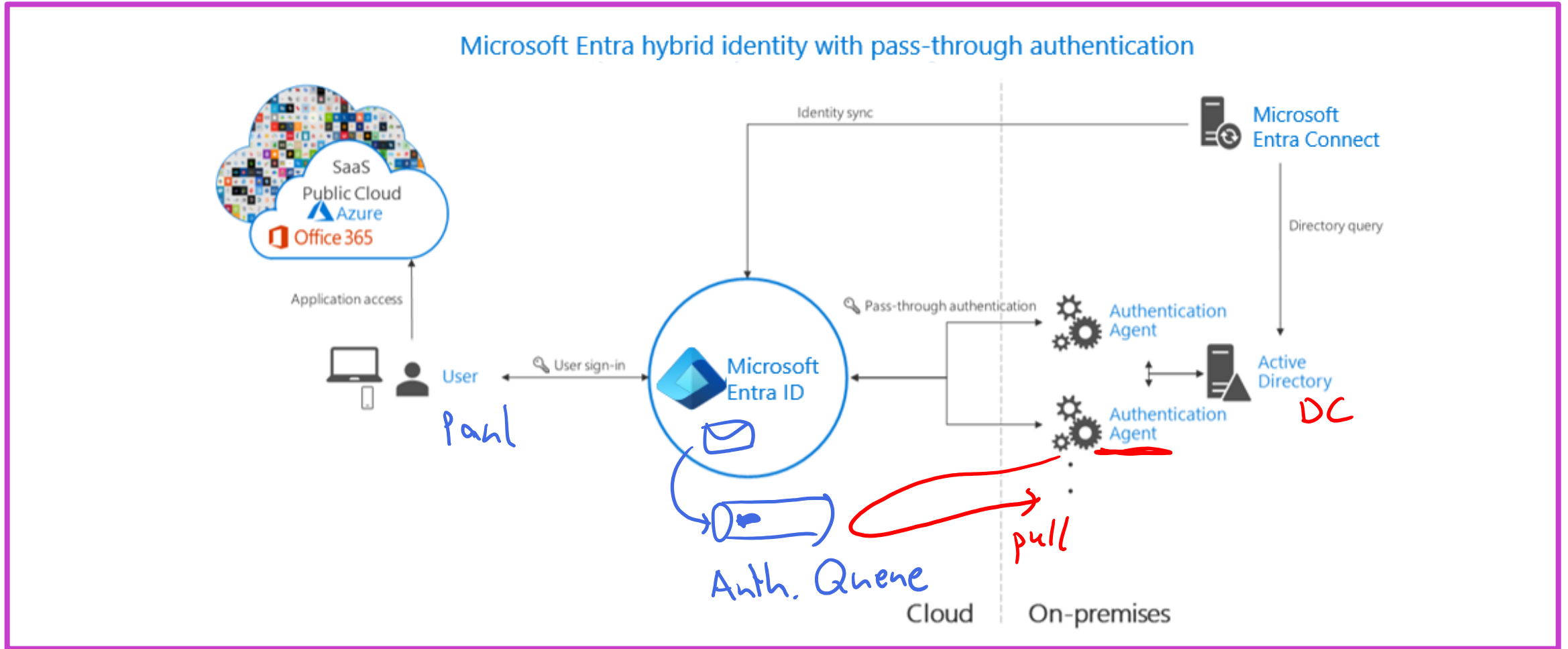
☐ Enable single sign-on ?

Previous Next

Implement and manage pass-through authentication (PTA)

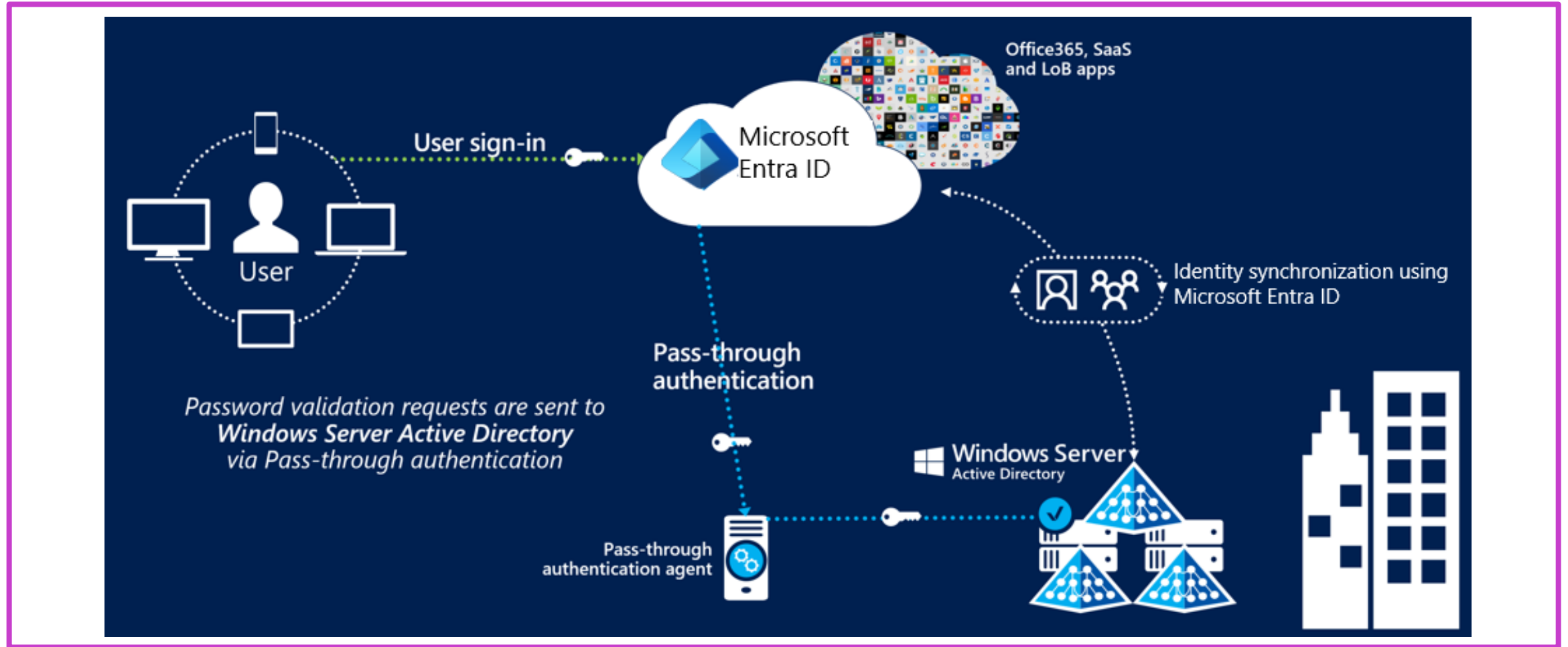


Pass-through authentication PTA



Agent requirements of pass-through authentication, using two agents for redundancy

How Microsoft Entra pass-through authentication works



Enable pass-through authentication

The screenshot shows the 'User sign-in' configuration window in the Microsoft Azure Active Directory Connect application. The left sidebar contains a navigation menu with the following items: Welcome, Express Settings, Required Components, User Sign-In (highlighted), Connect to Azure AD, Sync, Connect Directories, Azure AD sign-in, Domain/OU Filtering, Identifying users, Filtering, Optional Features, and Configure. The main content area is titled 'User sign-in' and includes the instruction 'Select the Sign On method.' Below this, there are four radio button options: Password Hash Synchronization, Pass-through authentication (which is selected), Federation with AD FS, Federation with PingFederate, and Do not configure. Each option has a help icon. Below the radio buttons, there is a section titled 'Select this option to enable single sign-on for your corporate desktop users:' with a checkbox labeled 'Enable single sign-on'. At the bottom of the main content area, there is a recommendation: 'We recommend that you have a cloud only Company Administrator account so that you are able to manage pass-through authentication in the event of an on-premises failure. Learn more'. The bottom of the window features 'Previous' and 'Next' buttons.

Microsoft Azure Active Directory Connect

Welcome
Express Settings
Required Components
User Sign-In
Connect to Azure AD
Sync
Connect Directories
Azure AD sign-in
Domain/OU Filtering
Identifying users
Filtering
Optional Features
Configure

User sign-in

Select the Sign On method.

- ☐ Password Hash Synchronization ?
- ☒ Pass-through authentication ?
- ☐ Federation with AD FS ?
- ☐ Federation with PingFederate ?
- ☐ Do not configure ?

Select this option to enable single sign-on for your corporate desktop users:

☐ Enable single sign-on ?

We recommend that you have a cloud only Company Administrator account so that you are able to manage pass-through authentication in the event of an on-premises failure. [Learn more](#)

Previous Next

The screenshot shows the 'Additional tasks' window in the Microsoft Azure Active Directory Connect application. The left sidebar is the same as the previous window, but the 'Tasks' item is highlighted. The main content area is titled 'Additional tasks' and includes the instruction 'The required tasks for the scenario have been completed. Choose from the list below to perform additional tasks.' Below this, there is a list of tasks: Privacy settings, View current configuration, Customize synchronization options, Configure device options, Refresh directory schema, Configure staging mode, Change user sign-in (highlighted), Configure Source Anchor, Manage federation, and Troubleshoot. Each task has a help icon. The bottom of the window features 'Previous' and 'Next' buttons.

Microsoft Azure Active Directory Connect

Welcome
Tasks

Additional tasks

The required tasks for the scenario have been completed. Choose from the list below to perform additional tasks.

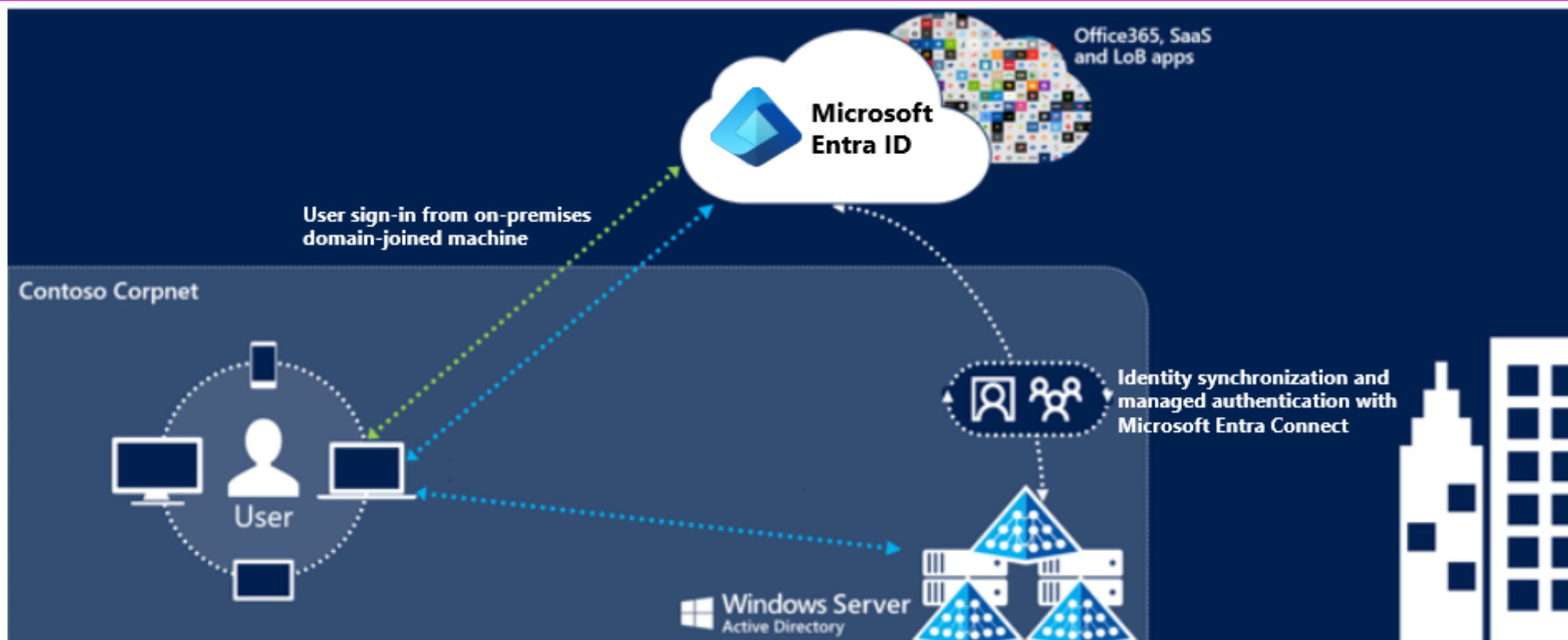
- Privacy settings
- View current configuration
- Customize synchronization options
- Configure device options ?
- Refresh directory schema
- Configure staging mode
- Change user sign-in**
- Configure Source Anchor
- Manage federation ?
- Troubleshoot

Previous Next

Seamless single sign-on (SSO)



Single sign-on



User sign-in via on-premises or via Microsoft Entra Connect alternate ID.

If single sign-in fails, the user is prompted to log in.

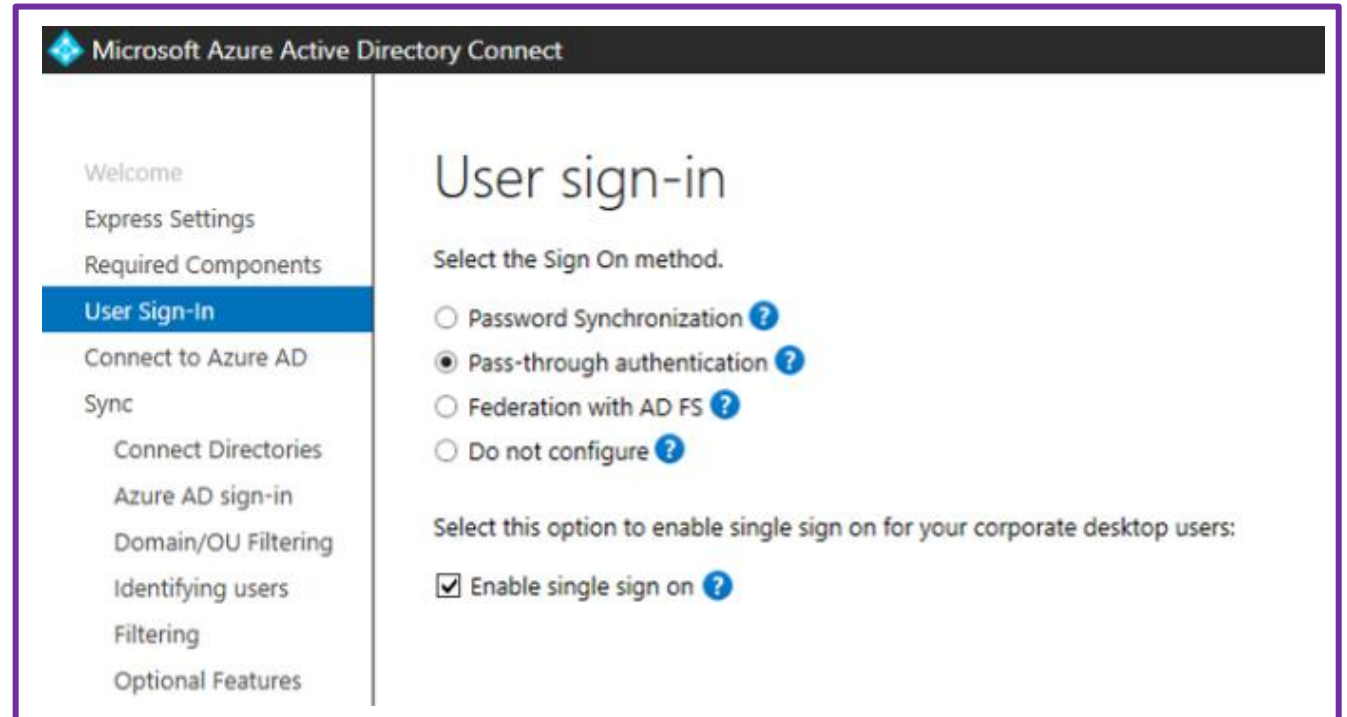
Sign out works, but releases the sign-in so all further access is blocked until sign-in is completed again.

Supported in browser-based and Office-based clients.

How seamless SSO is set up

Seamless SSO is enabled using Microsoft Entra Connect. While enabling the SSO, the following steps occur:

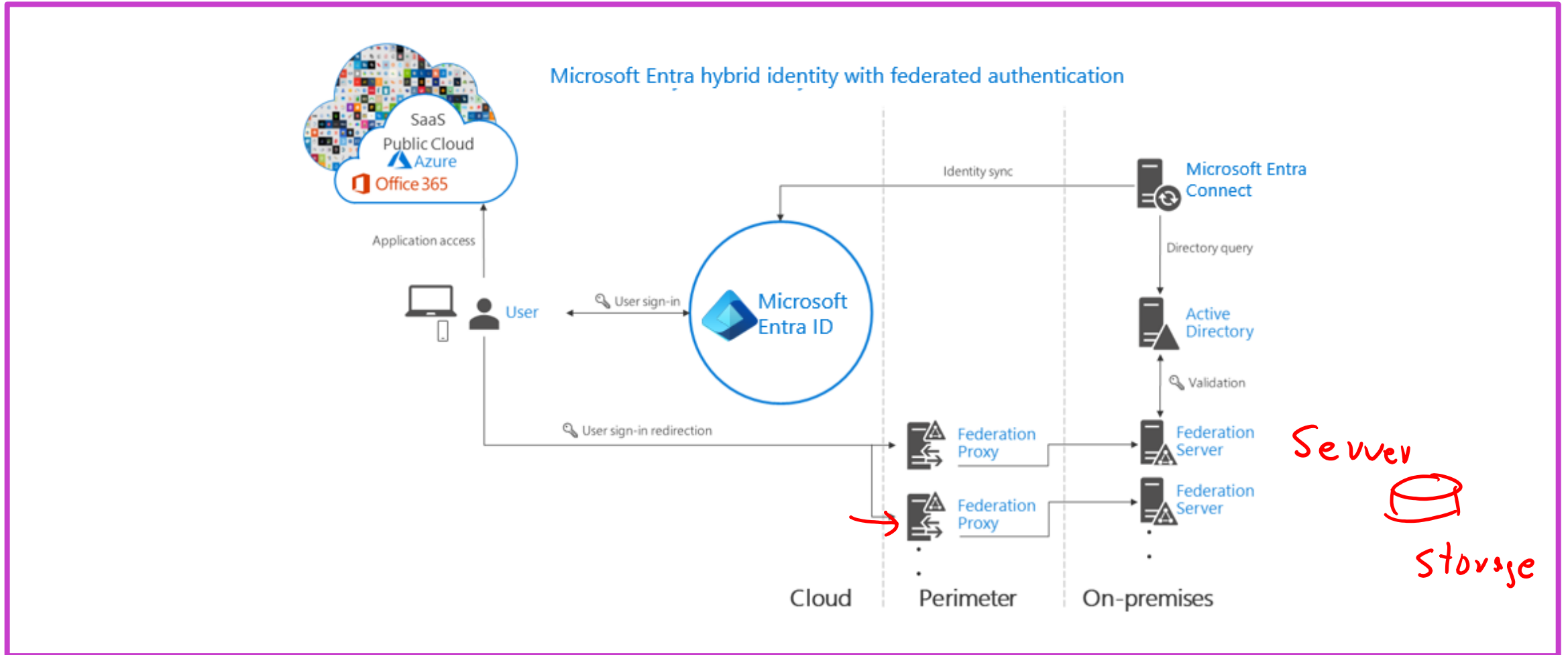
1. A computer account (AZUREADSSOACC) is created in your on-premises Active Directory (AD).
2. A set of Kerberos service principal names (SPNs) are created to be used during the Microsoft Entra ID sign-in process.
3. Kerberos decryption key is shared securely with Microsoft Entra ID.



Implement and manage federation



Federated authentication

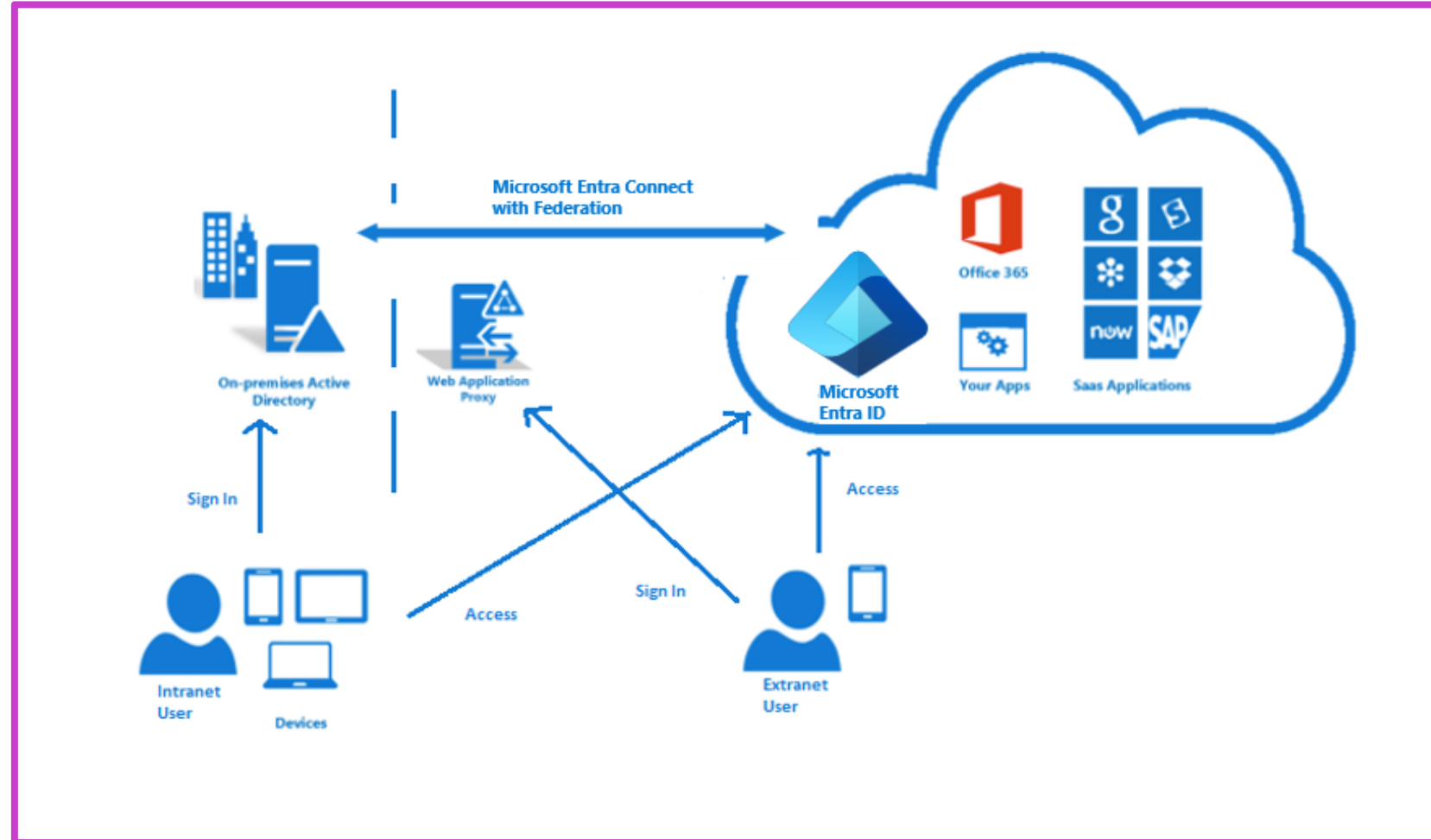


Components required for federation in the perimeter and internal network of your organization

What is federation?

Federation uses a new or existing farm with AD FS in Windows Server 2012 R2 and later

- Users sign in to Microsoft Entra ID services using their on-premises passwords
- Microsoft Entra Connect configures the trust between Microsoft Entra ID and the on-premises farm



Deploying a federation with AD FS and Microsoft Entra Connect

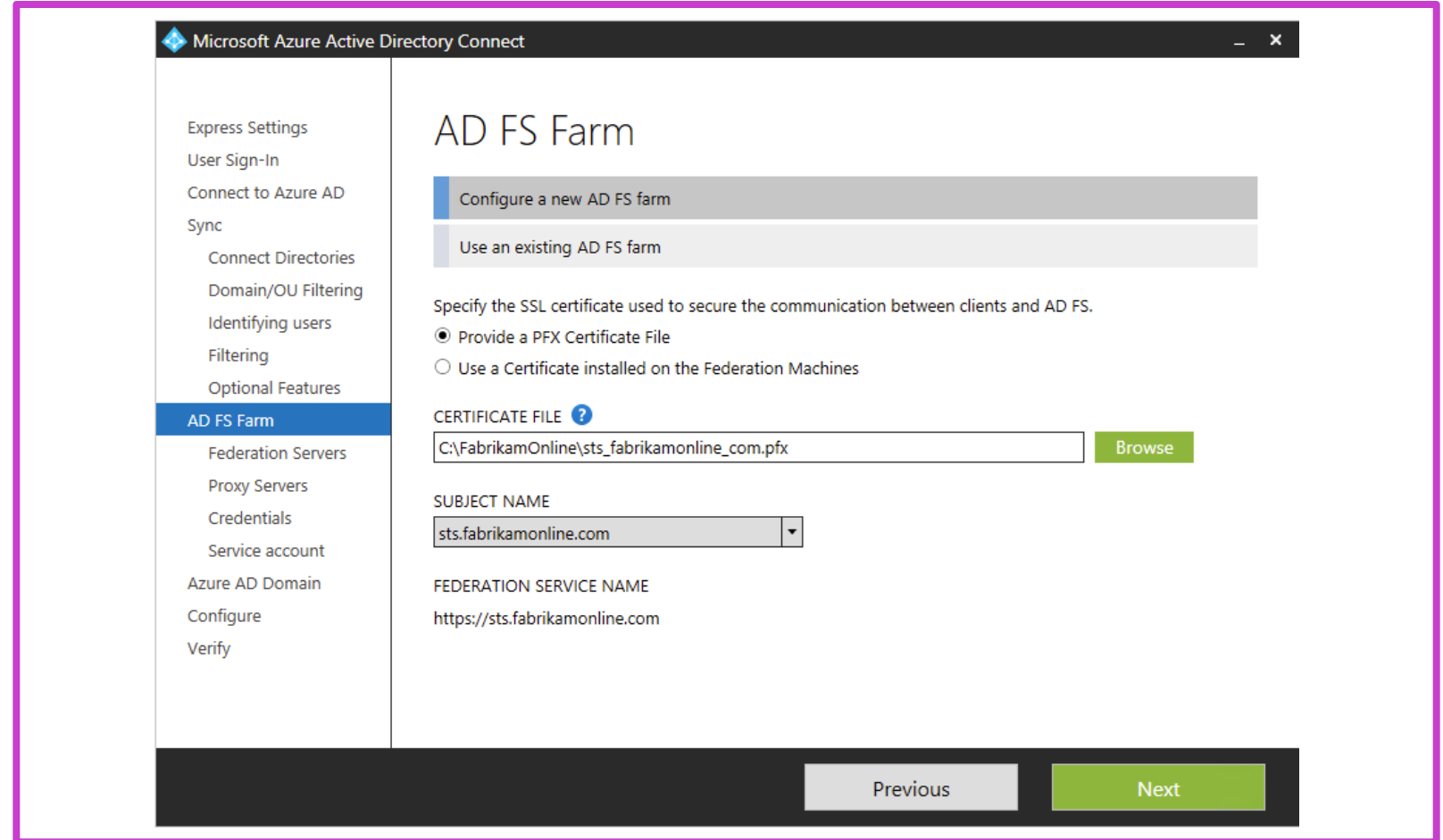
To deploy an AD FS farm, you need:

- Local administrator credentials on your federation servers.
- Local administrator credentials on any workgroup servers (not domain-joined) that you intend to deploy the Web Application Proxy role on.
- The machine that you run the wizard on to be able to connect to any other machines that you want to install AD FS or Web Application Proxy on by using Windows Remote Management.

Using Microsoft Entra Connect to connect to an AD FS farm

Primary steps

- Select AD FS server(s)
- Select Web App Proxy server(s)
- Specify AD FS service account
- Select Microsoft Entra domain to federate



The screenshot shows the 'Microsoft Azure Active Directory Connect' window with the 'AD FS Farm' configuration page. The left sidebar lists various settings, with 'AD FS Farm' selected. The main area has two options: 'Configure a new AD FS farm' (selected) and 'Use an existing AD FS farm'. Below these, there are radio buttons for 'Provide a PFX Certificate File' (selected) and 'Use a Certificate installed on the Federation Machines'. The 'CERTIFICATE FILE' section has a text box containing 'C:\FabrikamOnline\sts_fabrikamonline_com.pfx' and a 'Browse' button. The 'SUBJECT NAME' section has a dropdown menu showing 'sts.fabrikamonline.com'. The 'FEDERATION SERVICE NAME' section shows 'https://sts.fabrikamonline.com'. At the bottom, there are 'Previous' and 'Next' buttons.

Microsoft Azure Active Directory Connect

AD FS Farm

Configure a new AD FS farm

Use an existing AD FS farm

Specify the SSL certificate used to secure the communication between clients and AD FS.

☒ Provide a PFX Certificate File

☐ Use a Certificate installed on the Federation Machines

CERTIFICATE FILE ?

C:\FabrikamOnline\sts_fabrikamonline_com.pfx Browse

SUBJECT NAME

sts.fabrikamonline.com

FEDERATION SERVICE NAME

https://sts.fabrikamonline.com

Previous Next

Managing federation with Microsoft Entra Connect

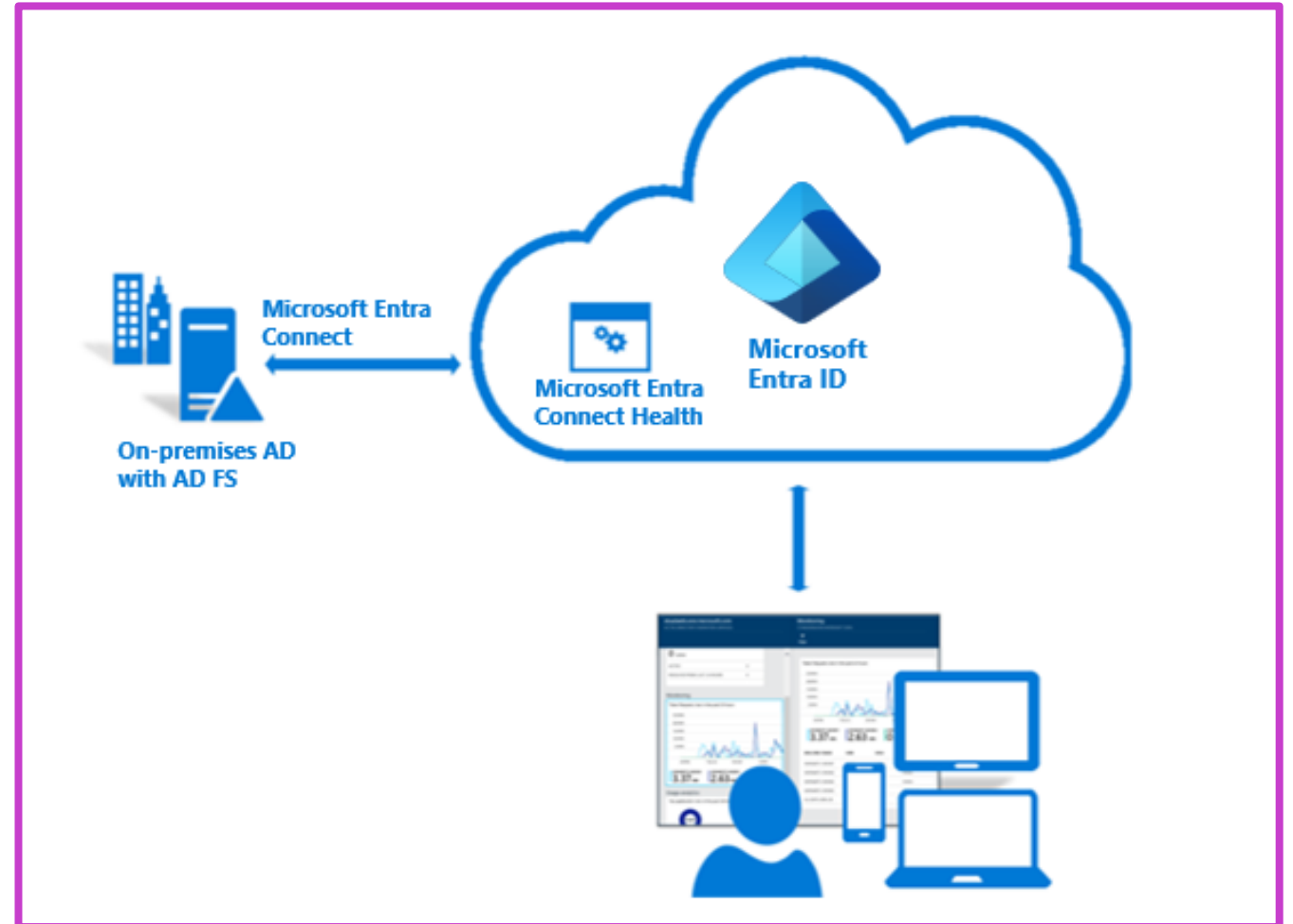
Manage AD FS feature	What it does
Repair the trust	How to repair the federation trust with Microsoft 365
Federate with Microsoft Entra ID using alternate login ID	Configure federation using alternate login ID
Add an AD FS server	How to expand an AD FS farm with an additional AD FS server
Add an AD FS Web Application Proxy <u>server</u>	How to expand an AD FS farm with an additional Web Application Proxy (WAP) server
Add a federated domain	How to add a federated domain

Implement Microsoft Entra Connect Health



What is Microsoft Entra Connect Health?

- Provides robust monitoring of your on-premises identity infrastructure
- Enables you to maintain a reliable connection to Microsoft 365 and Microsoft Online Services
- Provides monitoring capabilities for your key identity components
- Makes the key data points about these components easily accessible
- Microsoft Entra Health Connect portal (aka.ms/aadconnecthealth)

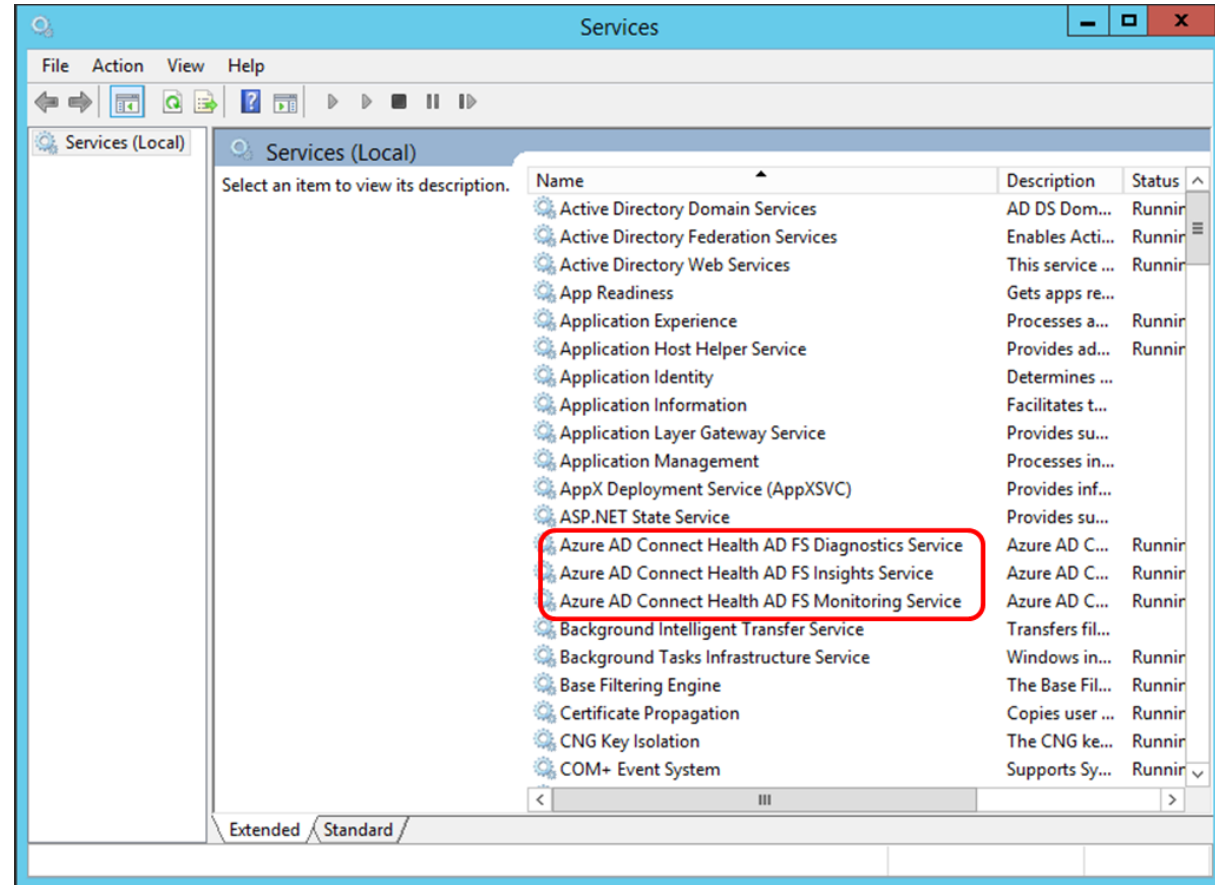


Considerations for installing Microsoft Entra Connect Health

- Microsoft Entra ID premium license required (P1 or P2)
- Only a global administrator can deploy and configure
- Microsoft Entra Connect Health Agent—install on each server to collect health data from
 - AD FS server or your Active Directory Domain Services machine
- Firewall must be opened for specific IP ranges and TCP ports
- TLS inspection can block install or operation
- PowerShell 4.0 or later required
- FIPS compliant encryption must be disabled

Example—Microsoft Entra Connect Help for AD FS installation steps

1. Install the agent for Active Directory Federation Service.
2. Configure the agent.
3. Sign in using a Microsoft Entra account with permission to register the agent.
4. Ensure that the Microsoft Entra Connect Health services are installed.
5. Install the Microsoft Entra Connect Health agent for Sync.



Manage Microsoft Entra Connect Health



Enable email notifications

The screenshot shows the Azure Active Directory Connect (Sync) Alerts page. The top navigation bar includes the URL 'fabtoso.onmicrosoft.com', a 'Delete' button, a 'Settings' button, and a 'Notification' section with 'Save' and 'Discard' buttons. The main content area is divided into three sections: 'Overview', 'Operations', and 'Alerts'. The 'Overview' section shows 'Azure Active Directory Connect Servers' with a count of 2, one 'Unhealthy' (FABVM03) and one 'Healthy' (FABVM02). The 'Operations' section shows 'Azure Active Directory Connect (Sync) Alerts' with 1 active alert. The 'Alerts' section shows a table with columns 'NAME', 'TYPE', and 'SCOPE'. A red box highlights the 'Notification Settings' icon in the top navigation bar. A blue banner at the top of the Alerts section states: 'You can provide feedback by doing a right click on any alert.'

Overview

Azure Active Directory Connect Servers

2

FABVM03	Unhealthy
FABVM02	Healthy

Operations

Azure Active Directory Connect (Sync) Alerts

1 active

Active	1
Resolved from last 24 hours	0

Alerts

Find ...

NAME	TYPE	SCOPE
ACTIVE ALERTS		
Azure AD Connect Sync Service is not r...	Error	FABVM03
RESOLVED ALERTS		
No items for this.		

Notification

fabtoso.onmicrosoft.com

Save Discard

Use the notification to get notified when there are new alerts.

OFF ON

Notify All Global Administrators ☐

ADDITIONAL EMAIL RECIPIENTS

varun@fabtoso.com

idadmins@fabtoso.com

Additional tasks

Delete a server or service instance

- Delete a server from the Microsoft Entra Connect Health service
- Delete a service instance from Microsoft Entra Connect Health service

Manage access with Azure role based access control

- Allow users or groups access to Microsoft Entra Connect Health
- Remove users or groups

Troubleshoot synchronization errors



Troubleshooting

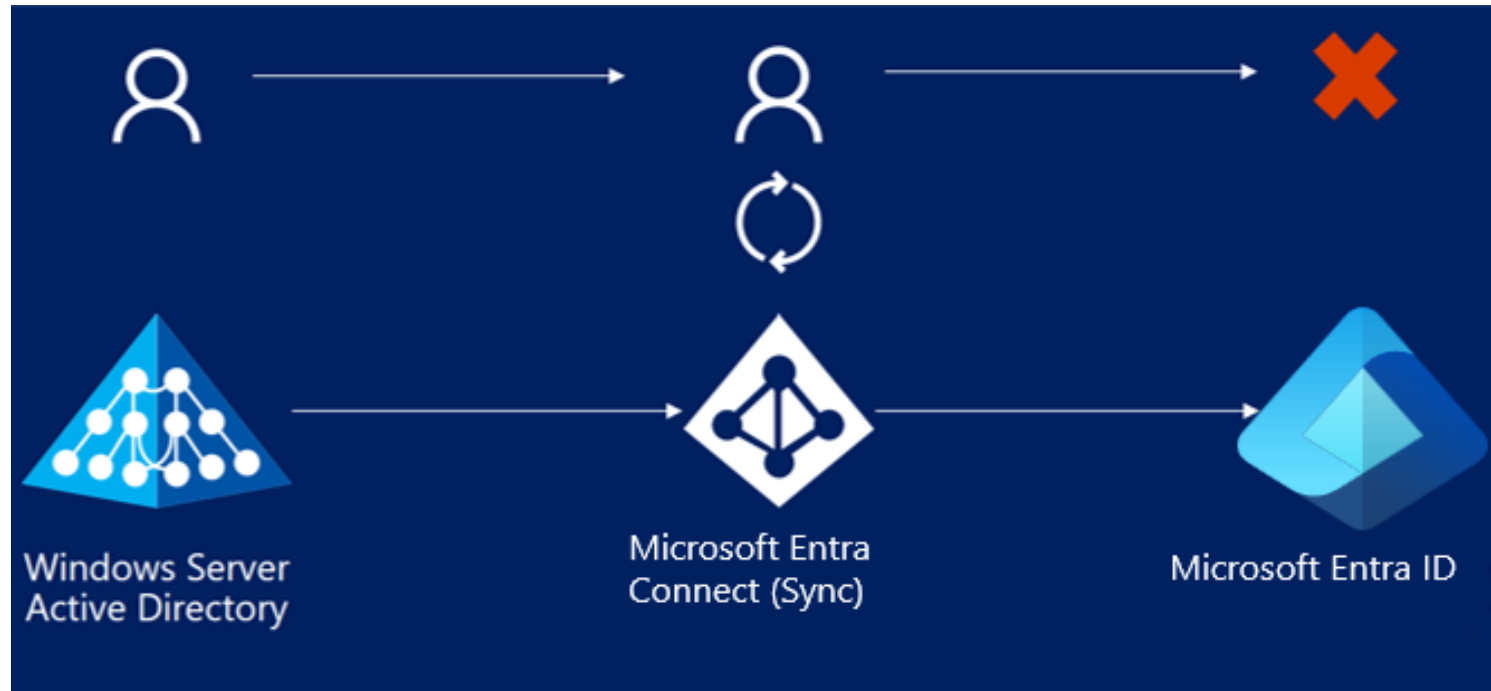
Diagnose and remediate duplicated attribute sync errors

- Duplicate UserPrincipalName or Proxy Addresses
- Orphaned object

Diagnostic and troubleshooting steps in Connect Health

- Attributes: UserPrincipalName, ProxyAddresses, SipProxyAddress, OnPremiseSecurityIdentifier

Potential synchronization errors



Data mismatch errors

- InvalidSoftMatch
- ObjectTypeMismatch

Duplicate attributes

- AttributeValueMustBeUnique

Data validation failures

- IdentityDataValidationFailed
- FederatedDomainChangeError

LargeObject

Admin role conflict

Summary

Initial Microsoft Entra ID configuration

- Microsoft Entra roles, custom roles **UAA**
- Custom domain
- Administrative units
- Tenant-wide settings

Configure and manage identities

- Users
- Groups
- Licenses
- Focus on proper maintenance!

External Identities

- Guests can use your solutions
- External collaboration
- Invite individually or in bulk
- Use external identity providers (as needed)

Hybrid Identity

- Connect to your on-premises AD
- Microsoft Entra Connect
- Implement PHS, PTA, and SSO
- Microsoft Entra Connect Health

Labs

Lab	Brief description	Length
1. Manage user roles	Create a user account, add a role to it, and remove a role	10 minutes
2. Working with tenant properties	Set tenant-wide properties such as changing the tenant display name and adding privacy information	10 minutes
3. Assign licenses to users and groups	Create a user in Microsoft Entra ID, create a security group, and assign a license to a group	10 minutes
4. Configure external collaboration	Configure external collaboration settings	5 minutes
5. Add guest users to the directory	Add users individually to the directory	5 minutes
6. Add a federated identity provider	Configure a new federated identity provider	15 minutes

References

Implement initial configuration of Microsoft Entra ID

<https://learn.microsoft.com/learn/modules/implement-initial-configuration-of-azure-active-directory/>

Create, configure, and manage identities

<https://learn.microsoft.com/learn/modules/create-configure-manage-identities/>

Implement and manage external identities

<https://learn.microsoft.com/learn/modules/implement-manage-external-identities/>

Implement and manage hybrid identity

<https://learn.microsoft.com/learn/modules/implement-manage-hybrid-identity/>



Learning path recap

In this learning path, we:

Explained and configured identities for use in your cloud solutions.

Deployed and managed a Microsoft Entra ID infrastructure.

Configured and maintained external and hybrid identities.

